



TECHNICAL REPORT

TEAM #06

ECLIFT

30/06/2024



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Contents

1.	Introduction	6
1.1.	Abstract	6
1.2.	The XtraChallenge.....	6
1.3.	The École Centrale Lyon and ECLift	6
2.	Project Management	6
2.1.	Team Structure.....	6
2.2.	Development Masterplan	7
2.3.	Sponsors.....	8
2.4.	Project Value	8
3.	Electronics and Propulsion.....	9
3.1.	Propeller Selection	9
3.1.1.	Theoretical Analysis.....	9
3.1.2.	Experimental Analysis	10
3.2.	Electronic components	10
3.2.1.	Electronic Speed Controller	10
3.2.2.	Main Battery	11
3.2.3.	Servomotors	12
3.2.4.	Auxiliary Battery.....	13
3.2.5.	Radio Transmitter and Receiver (TX-RX)	13
3.2.6.	Voltage Regulator	14
3.3.	Circuit.....	14
4.	Aerodynamic Design.....	15
4.1.	Airfoil Design.....	15
4.2.	Wing Design	16
4.3.	Winglet Design	17
4.4.	Tail Design.....	19
4.4.1.	Horizontal Stabilizer	19
4.4.2.	Vertical Stabilizer	20
4.5.	Aerodynamic analysis	20
5.	Stability and Control.....	21
5.1.	Stability analysis.....	21

5.2.	Control surfaces	23
6.	Structural Design	24
6.1.	Introduction	24
6.2.	Fuselage Design	24
6.2.1.	Main Body	25
6.2.2.	Tail Body	26
6.3.	Wing Design	27
6.3.1.	Inner Wing Design	27
6.3.2.	Outer Wing Design	28
6.4.	Empennage Design.....	29
6.5.	Center of Mass vs Payload	29
6.6.	Structure Simulation.....	30
6.7.	Box Design.....	31
7.	Manufacturing	32
7.1.	Preparation	32
7.2.	Fuselage Assembly	33
7.3.	Wing Assembly	34
7.4.	Assembled Aircraft	35
8.	Aircraft Performance	36
9.	Innovations.....	38
10.	Lessons learned	38
11.	Closure.....	39
	References.....	40
	Annex I: Budget Breakdown	41
	Annex II: Aircraft Drawings.....	43

Table of Figures

Figure 1: Organization and hierarchy of the team during the 23/24 season.....	7
Figure 2: ECLifter Development Masterplan.....	7
Figure 3: Logos of ECLifter's sponsors	8
Figure 4: Thrust model for both propellers.	9
Figure 5: Efficiency evolution with flight speed.....	10
Figure 6: Flight Circuit	12
Figure 7: Torque required to deflect control surfaces	13
Figure 8: Electronic components layout	14
Figure 9: Analysis of Airfoil Performance Metrics at $Re=200,000$	15
Figure 10: S1221 airfoil.....	16
Figure 11: Initial Wing Geometry	17
Figure 12: Final Wing Planform.....	17
Figure 13: Comparative Analysis of Winglet Types	18
Figure 14: Sharklet winglet.....	19
Figure 15: Empennage Design	20
Figure 16: CFD study of the full aircraft	20
Figure 17: Longitudinal stability - C_m vs α	22
Figure 18: Dynamic stability root locus.....	22
Figure 19: An exploded view of the main body fuselage.....	25
Figure 20: An exploded view of the tail body fuselage.	27
Figure 21: An exploded view of the inboard wing.	28
Figure 22: An exploded view of the outboard wing.	28
Figure 23: Exploded Views of the Rudder and Elevator	29
Figure 24: Distance to the C.G. from the L.E. of the wing as a function of billiard balls.	30
Figure 25: Landing gear under static loading in COMSOL	30
Figure 26: Top view of the wing under static loading in COMSOL	31
Figure 27: ECLifter's Box, made of wood.....	31
Figure 28: The .DXF file of the main body fuselage.....	32
Figure 29: The internal structure of the first prototype ECLifter	33
Figure 30: Manufacturing of the revised fuselage structure.....	33

Figure 31: The 3mm balsa is being fitted to the leading edge of the wing. On top is the inboard section, on bottom the outboard.	35
Figure 32: Render of the assembled ECLifter.....	35
Figure 33: Aircraft Polar	36
Figure 34: Payload Prediction as a function of air density.....	37
Figure 35: Aftermath of the second flight of the ECLifter	38
Figure 36: Final approach during a test flight on June 30th, 2024.....	39

Table of Symbols

SYMBOL	PARAMETER	UNITS
$ q _{bp}$	Battery Capacity	[mAh]
I_{max}	Electric Current	[A]
T_{servo}	Torque of the servo	[N]
V	Velocity	[m/s]
c_f	Chord of control surface	[m]
b_f	Length of control surface	[m]
δ_f	Deflection of control surface	[°]
θ_f	Operational range of servo	[°]
L/D	Lift to drag ratio	[-]
C_L	Lift Coefficient	[-]
C_D	Drag Coefficient	[-]
C_M	Momentum Coefficient	[-]
Re	Reynolds Number	[-]
MAC	Mean aerodynamic chord	[-]
b_w	Wingspan	[-]

S_w	Wing surface	[-]
λ_w	Taper Ratio	[-]
C _{Ma}	Momentum coefficient with the angle of attack	[-]
C _{Lb}	Rolling momentum coefficient with the side slip angle	[-]
C _{Nb}	Yawing momentum coefficient with the side slip angle	[-]
S.M.	Static margin	[%]
x_{NP}	Neutral point position	[m]
x_{CG}	Position of center of gravity	[m]
MTOW	Maximum take-off weight	[kg]
OEW	Operational empty weight	[kg]
C _m δ_e	Elevator efficiency	[-]
C _L δ_a	Aileron efficiency	[-]
C _N δ_a	Induced yaw	[-]
C _N δ_R	Rudder efficiency	[-]
C _L δ_R	Induced roll	[-]
ρ	Density	$\left[\frac{kg}{m^3}\right]$
MPL	Maximum Payload	[kg]

1. Introduction

1.1. Abstract

The purpose of this Technical Report is to present a comprehensive engineering document containing all the development process of the ECLifter: the airplane that will represent ECL in the upcoming XtraChallenge 2024. It contains an introduction to the project itself, and an overview of all the different scientific tools used to develop the aircraft from an aerodynamic, structural, and electronic standpoint.

1.2. The XtraChallenge

The XtraChallenge is an engineering competition that takes place every year in Valencia (Spain) since 2023. It is organized by the Xtra2 UPV team, recognized for being the best Spanish team to ever compete at the renowned Air Cargo Challenge. In this competition, aeronautics students' teams, mostly coming from Spanish national territory, gather to present and fly original airplanes made by themselves in every aspect, following the restrictions of a series of regulations that change every year.

1.3. The École Centrale Lyon and ECLift

École Centrale Lyon is a French engineering university dating from 1857, located in Lyon. ECLift (read: “easy lift”) is a group of students from the Master’s in Aerospace Engineering program offered by this school. ECLift’s activity began in October 2023, when thirteen members coming from seven different nations started working on the development on an aircraft for the XtraChallenge. Ten of those members will travel to Valencia in July of 2024 to participate in the contest, and to showcase their prototype in front of top-level competitors, academics, and companies.

2. Project Management

2.1. Team Structure

The team is structured in different departments, separating:

- **Management and Logistics**, which takes care of procuring materials, meeting with stakeholders, and directing the other departments and their components.
- **Aerodynamic Design and Simulation**, in charge of the study of different airfoils, wing planforms, stability criteria, and other topics regarding the geometry choice.
- **Structures and Manufacturing**, which designs, sizes, and validates all of the architectural elements of the airframe, to be able to fabricate them afterwards.
- **Propulsion and Onboard Electronics**, in charge of the powertrain study: thrust optimization, choice of electronic components, and the circuit layout.

A head for each department was established at the beginning of the project.

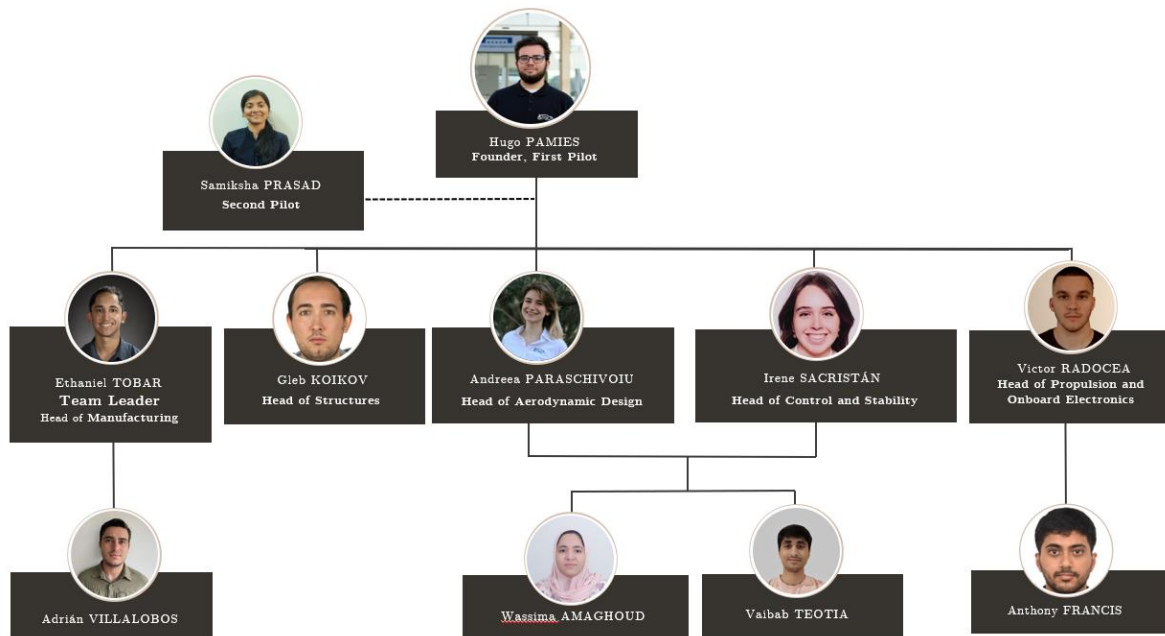


Figure 1: Organization and hierarchy of the team during the 23/24 season

2.2. Development Masterplan

The development of the season followed a Masterplan established in early October by the Management and Logistics department, in charge of the production of a reasonable Gantt chart. A simplified version is presented here.

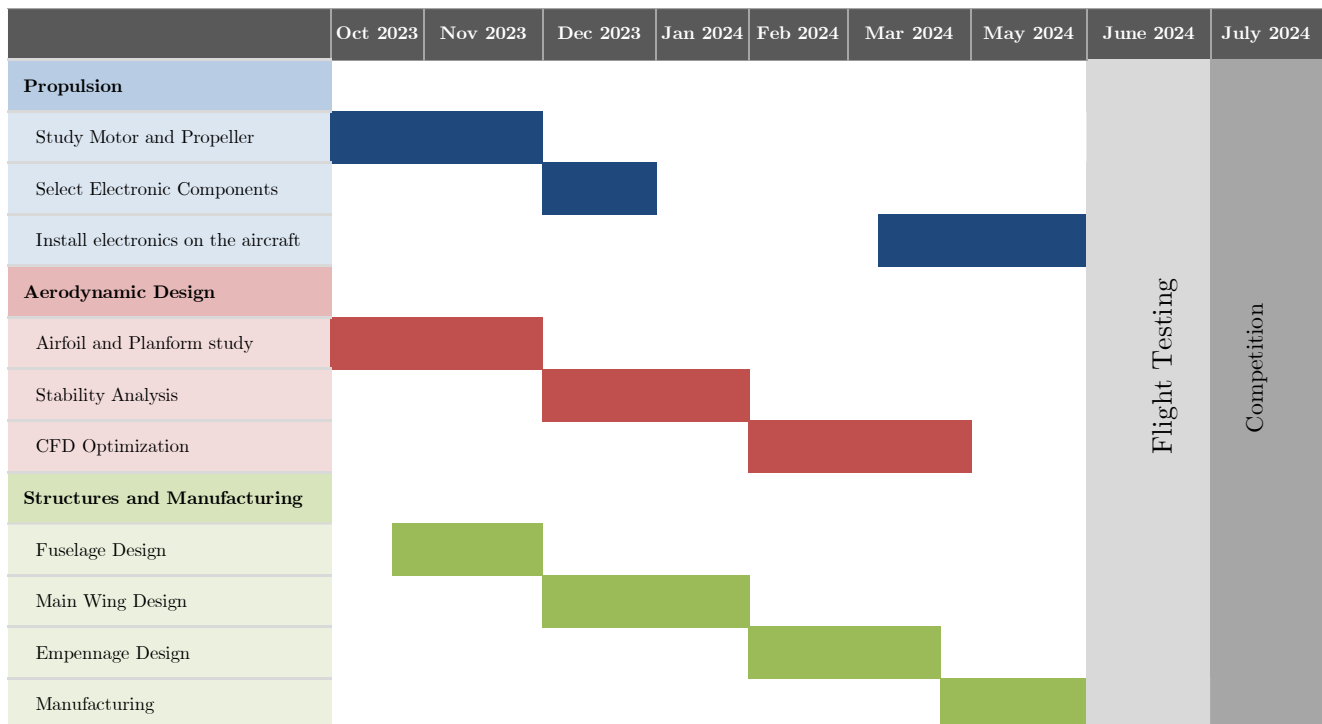


Figure 2: ECLifter Development Masterplan

2.3. Sponsors

The main sponsors and collaborators of ECLift are:

- **École Centrale de Lyon** is a prestigious French engineering school, renowned for its rigorous curriculum in science and engineering, preparing students for high-level careers in industry and research.
- **Laboratoire de Tribologie et Dynamique des Systèmes (LTDS)** is a French research laboratory focusing on tribology, the study of friction, wear, and lubrication, and the dynamics of mechanical systems.
- **Centrale Innovation** is a hub for technological development and innovation, fostering collaboration between research, academia, and industry to drive advanced solutions and promote scientific and technological progress.
- **Safran Aircraft Engines** is a leading global company that designs, manufactures, and maintains aircraft engines for commercial and military aviation, renowned for innovation and technological excellence.
- **Thales** is a multinational company specializing in aerospace, defense, security, and transportation, known for innovative technologies in cybersecurity, AI, and IoT.



Figure 3: Logos of ECLift's sponsors

2.4. Project Value

In this section, the economic value of the whole project is divided into different categories:

CATEGORY	COST
CONSTRUCTION MATERIALS	294,40 €
ELECTRONICS AND POWERTRAIN	450,90 €
CONSUMABLES	117,00 €
ACCESSORIES	35,20 €
PARTICIPATION COSTS	2.240,00 €
TOTAL PROJECT VALUE	3.137,50 €

Table 1: Project Value

The most relevant expense being the participation costs, including registration and travel. The complete description of the entire budget can be found in Annex I: Budget Breakdown.

3. Electronics and Propulsion

This section describes all the electronic components installed on the aircraft which constitute both the flight control system and the propulsion system. In accordance with the competition's regulations, some of the component choices were limited to a given manufacturer's brand and model and some others were limited in terms of specifications.

The strictly imposed components are the electric motor, Emax GT2820/06 985KV, and the propeller, either APC 10x6E or Aeronaut CamCarbon Light 10"x6". The components for which the limitations are imposed in terms of performance are the main battery, which should have 3 cells in series and not exceed 12.6V, and the ESC, which should allow for a constant current of 30A.

3.1. Propeller Selection

Considering the limitations of the competition, the highest thrust can only be obtained by figuring out which of the two available propellers offers the best performance when paired with the imposed electric motor. Finding the optimal pair is a critical aspect of the design process since the obtained thrust will influence the choices made in the plane's design and strategy. The optimization process was performed both theoretically, using MotoCalc software, and also through experimental testing using a dedicated L-arm static thrust measuring stand, which was in-house developed.

The two propellers are both 10x6, lightweight propellers capable of producing high thrust. The Aeronaut CAMcarbon Light 10"x6" propeller is made from carbon fiber, while the APC 10x6E is made from a lightweight yet also resistant plastic. Despite the difference in materials, both propellers have similar weights below 30g.

3.1.1. Theoretical Analysis

The MotoCalc software showed us that the difference between the two propellers is less than 10% in terms of thrust, but the highest value is obtained with the Aeronaut CamCarbon Light 10"x6" propeller and is about 1200g of thrust, as shown in Figure 4.

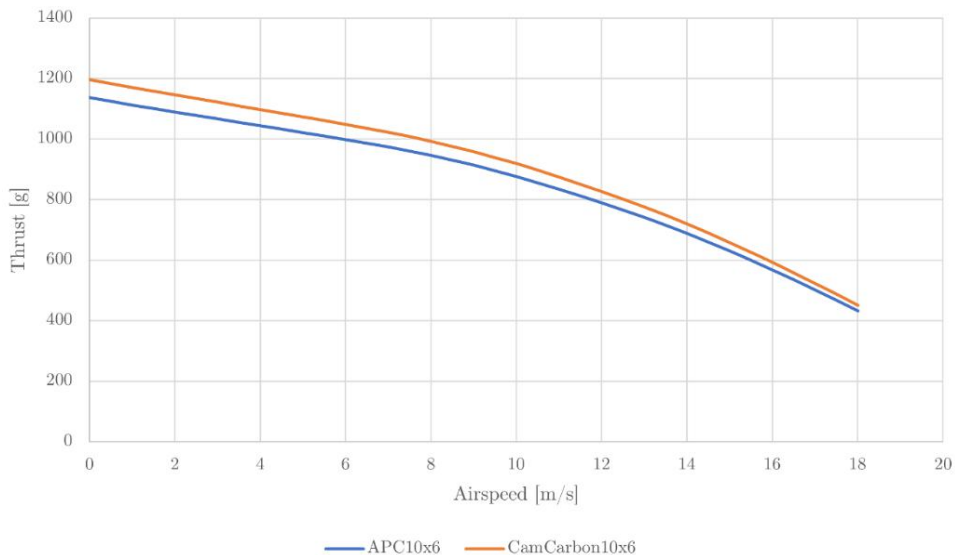


Figure 4: Thrust model for both propellers.

By analyzing the efficiencies, we observe that the CAM Carbon 10x6 propeller is slightly more efficient on the entire flight airspeed range compared to the APC 10x6, but the difference is less than 3%. This is shown in Figure 5, which presents the produced thrust per input Watt.

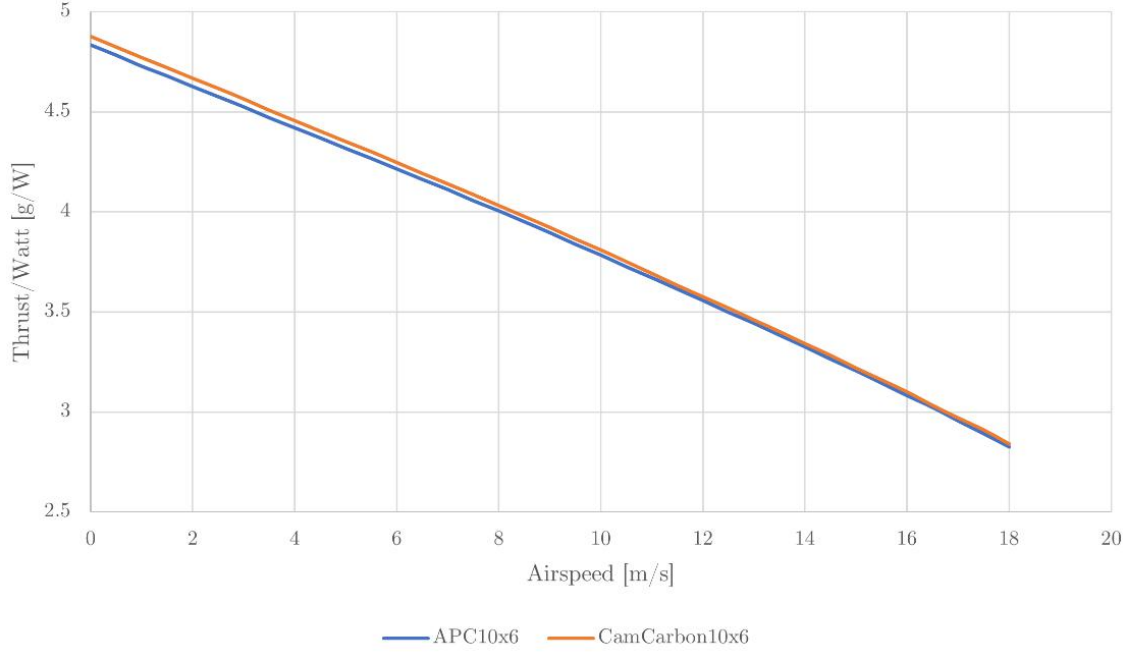


Figure 5: Efficiency evolution with flight speed

Because the predicted difference in thrust is less than 10%, and the differences in propeller efficiency and gram of thrust per input Watt are minimal, the selected type of propeller was the APC 10x6 propeller because of the reduced price and increased availability compared to the CAMcarbon 10x6 propeller.

3.1.2. Experimental Analysis

To validate the theoretical results, the electric motor, paired with the APC 10x6 propeller were mounted in the in-house developed static thrust testing stand. The motor run-up test was performed up to 100% throttle and the values of the thrust were registered using a flat-bottom scale installed at the lower end of the L-shaped stand. At the maximum throttle setting, the experiment registered a value of 1079g of thrust, slightly lower than the estimated 1137g from the theoretical analysis.

3.2. Electronic components

3.2.1. Electronic Speed Controller

An effective speed controller was selected while complying with the regulation that it supports a constant current of at least 30A. SKYWALKER 50A V2 was selected as it delivers better performance when compared with other models of a similar price range.

FEATURE	SKYWALKER 50A V2	FLYFUN 40A BEC	TURNIGY PLUSH 60A	EMAX BLHELI SERIES 40A
CURRENT CAPACITY	50A continuous current, providing higher power capacity	40A continuous current, suitable for many competitions setups	60A continuous current, higher capacity but might be overkill for some setups	40A continuous current, sufficient but less powerful
INTEGRATED BEC	Not integrated	Built-in BEC providing stable voltage to receiver and servos	Built-in BEC, but less praised for stability	Not integrated
FIRMWARE	Advanced firmware with improved throttle response and smoother motor control	Smooth and precise throttle control	Programmable but not as user-friendly	BLHeli firmware, good but not as refined for high-level competition
DURABILITY	Built to withstand high currents and rapid throttle changes, making it more durable for competition	Designed with thermal protection for safety	Durable, but heavier and bulkier	Durable, but not as robust under extreme conditions
PROGRAMMABILITY	Extensive programmability for optimal performance	Highly programmable with customizable settings	Limited customization options	Limited programmability compared to others

Table 2: ESC feature matrix

3.2.2. Main Battery

The main battery was considered based on the flight circuit requirement and based on regulations as well. The initial step in accurately sizing the battery involves considering the following mission times:

- 10 seconds of take-off.
- 30 seconds of climb.
- 180 seconds of cruise (with a margin for missed checkpoints).
- 80 seconds for landing (accounting for multiple approaches).

$$|q|_{bp} = c \cdot t \cdot I_{max} = 1.2 \cdot \frac{300}{3600} \cdot 30000 = 3000 \text{ mAh}$$

Taking a safety coefficient of 1.2 and considering the total flight duration to be 300 seconds (4.33 minutes), the battery required to complete the flight needs a capacity of 3000 mAh.

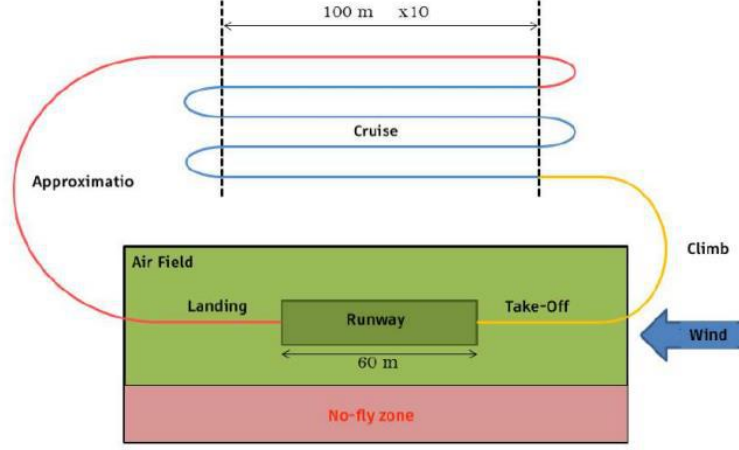


Figure 6: Flight Circuit

Based on the previous calculations for the minimum requirement of energy and also considered a safety factor, the ACCU LIPO SLS XTRON 3200mAh 3S 20C battery is selected as it offers several advantages over other batteries. With a capacity of 3200mAh, it provides a balance between power and weight, ensuring longer flight times without significantly increasing the aircraft's weight. The real capacity of the battery is usually lower than advertised, therefore we are still in the working range.

Compared to other brands, SLS (Stark Light Systems) batteries are known for their superior quality and reliability. They have lower internal resistance, resulting in less voltage drop under load. The voltage drop in the SLS XTRON 3200mAh 3S 20C is minimal, ensuring that our plane maintains power efficiency throughout the flight. This stability is critical in competitions where consistent performance can make a significant difference. Therefore, choosing the SLS XTRON battery over other options ensures better power management, longer flight times, and reliable performance, making it a superior choice for competitive flying.

3.2.3. Servomotors

While choosing the servomotors, our main goal is to select components with a minimum weight and size, so that they can be integrated easily and seamlessly integrated in the wing structures, tail and empennage, while also being powerful enough to withstand and overcome the aerodynamic forces applied on the command surfaces in order to ensure that the plane can be controlled.

In order to estimate the torque applied on all the control surfaces, the following formula was used:

$$T_{servo}(V) = 3.06 \cdot 10^{-6} \cdot \frac{V^2 \cdot c_f^2 \cdot b_f \cdot \sin(\delta_f) \cdot \tan(\delta_f)}{\tan(\theta_{servo})}$$

Where: c_f is the chord of the control surface, in cm; b_f is the length of the control surface, in cm; V is the airspeed, in mph; δ_f is the deflection of the given surface, in deg; θ_{servo} is the operational range of the servo from the neutral point, in degrees.

For a variation of the velocity from $[0 \div 20]$ m/s, the variation of the torque is presented in Figure 7.

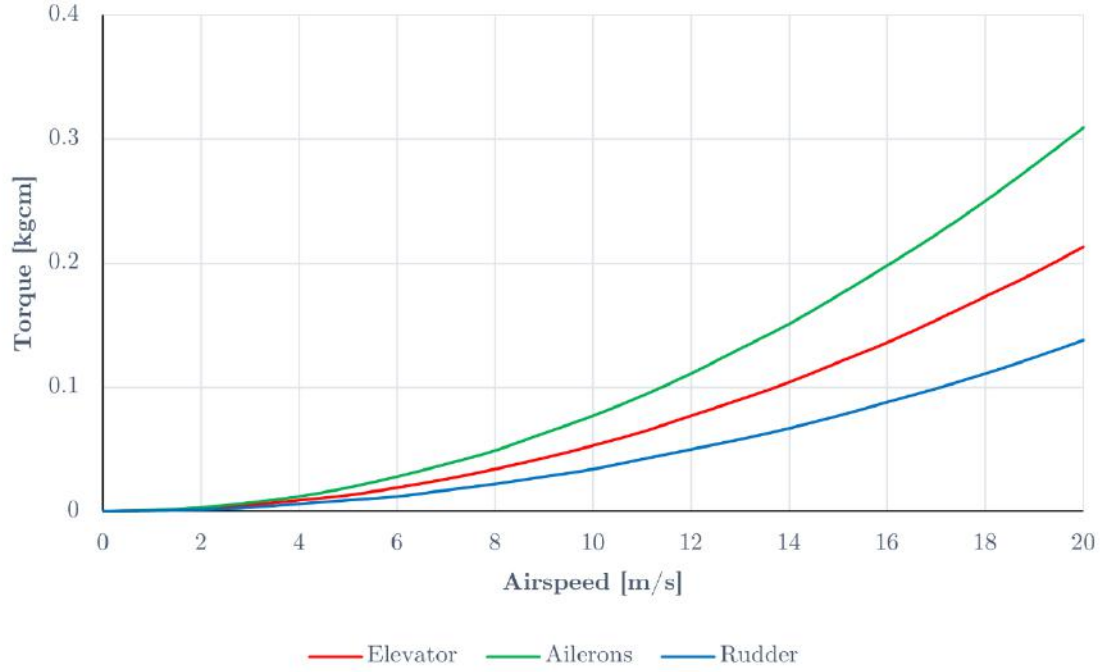


Figure 7: Torque required to deflect control surfaces

From this figure, we can observe that the most solicited control surfaces are the ailerons. Thus, the servomotors need to be strong enough to resist these aerodynamic forces.

For our targeted flight speed of around 15 m/s, we can observe that the maximum torque to which the servos need to resist is about 0.2 kg·cm; thus, we selected the EMAX ES3004 17grs/3.5kg·cm servomotor, which satisfies by far the strength criterion, while also being compact and relatively lightweight, making it very easy to integrate into both the wings and empennage. Another reason to select it was the fact that it is one of the cheapest full metal gear servos in the market. A feature that was considered a must for reliability reasons.

3.2.4. Auxiliary Battery

An Accu Lipo 2S 45C Black Lithium battery with a nominal capacity of 500mAh was chosen to power the flight controller, servos, and receiver. It is suitable since these components do not demand a significant amount of power, and it enables the isolation of the control circuit from power electronics for the safety reasons stated in the regulations.

3.2.5. Radio Transmitter and Receiver (TX-RX)

The Radio FS-i6 + Receiver FS-IA6 FLYSKY is selected as it is an excellent choice due to its reliable 2.4 GHz frequency operation, essential safety features like the programmable power cut-off lever, robust signal integrity with AFHDS 2A protocol, user-friendly interface, and affordability. These features ensure compliance with competition regulations, enhance safety, and provide a reliable and responsive control system, making it a superior option for competitive flying.

3.2.6. Voltage Regulator

Using the UBEC 4A Regulator for the auxiliary battery unit ensures a stable and efficient power supply for the plane's electronics. Its high current capacity and efficiency keeps the components safe. This regulator helps maintain consistent power and protects against electrical issues. A switch for ON/OFF the control circuit as well has been incorporated.

3.3. Circuit

The above-mentioned components are installed inside the body of the aircraft as presented in Figure 8.

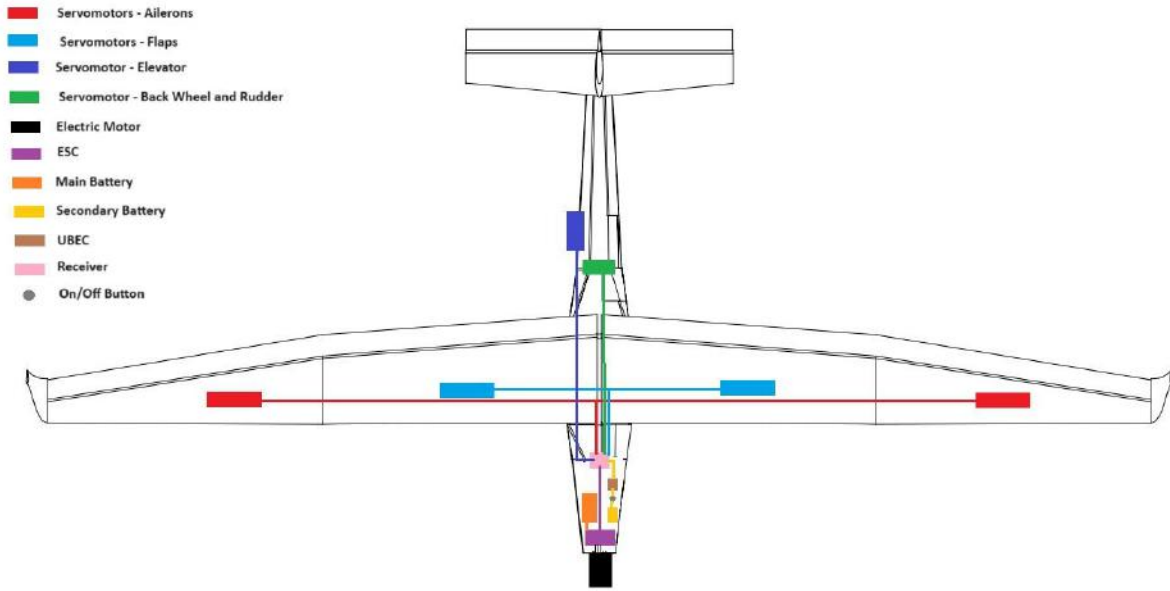


Figure 8: Electronic components layout

Inside each half-wing we can find two servomotors which actuate both the ailerons and the flaps. At the tail section we can find two other servomotors which control the elevator and rudder and tail wheel.

The electric motor is out-running, thus is installed outside the fuselage, in the nose section and is subjected to direct airflow for cooling. Also, in the nose section in the fuselage we can find stored all the other electronic components used to control the aircraft, the ESC, main and secondary batteries, the UBEC and the receiver.

In order to effectively turn on and off the airplane, in between the secondary battery and the UBEC, a two-positions button has been installed. Connections are ensured by lightweight cables which are neatly fixed to the inside of the airframe following the shortest possible path. Connectors grant easy disassembly of the aircraft without entanglement problems.

4. Aerodynamic Design

This section delves into the critical components of aerodynamic design, including the airfoil, wing, winglet, and tail configurations, each playing a vital role in optimizing flight characteristics. Airfoil selection focuses on maximizing lift and minimizing drag, while wing design emphasizes span, chord distribution, and taper ratio for optimal lift-to-drag balance. Winglets are analyzed for their effectiveness in reducing induced drag, and tail design ensures stability and control. The section also provides comprehensive aerodynamic analysis, including CFD simulations, which supports the refinement of these elements to enhance overall aircraft performance.

4.1. Airfoil Design

The aerodynamic performance of an aircraft is critically dependent on the design of its airfoil. Our approach commenced with a comprehensive analysis of various airfoil profiles, evaluating each based on key aerodynamic metrics, such as the lift-to-drag ratio (L/D), lift coefficient (C_l), and moment coefficient (C_m). For the initial iteration, a Reynolds number (Re) of 200,000 was assumed. This process led to the identification of seven promising airfoils. The use of XFLR5 facilitated an in-depth evaluation of these parameters across different conditions, yielding a robust dataset for comparative analysis. The studied airfoils were in the range of [10,12.5] % thickness and [5,8] % camber.

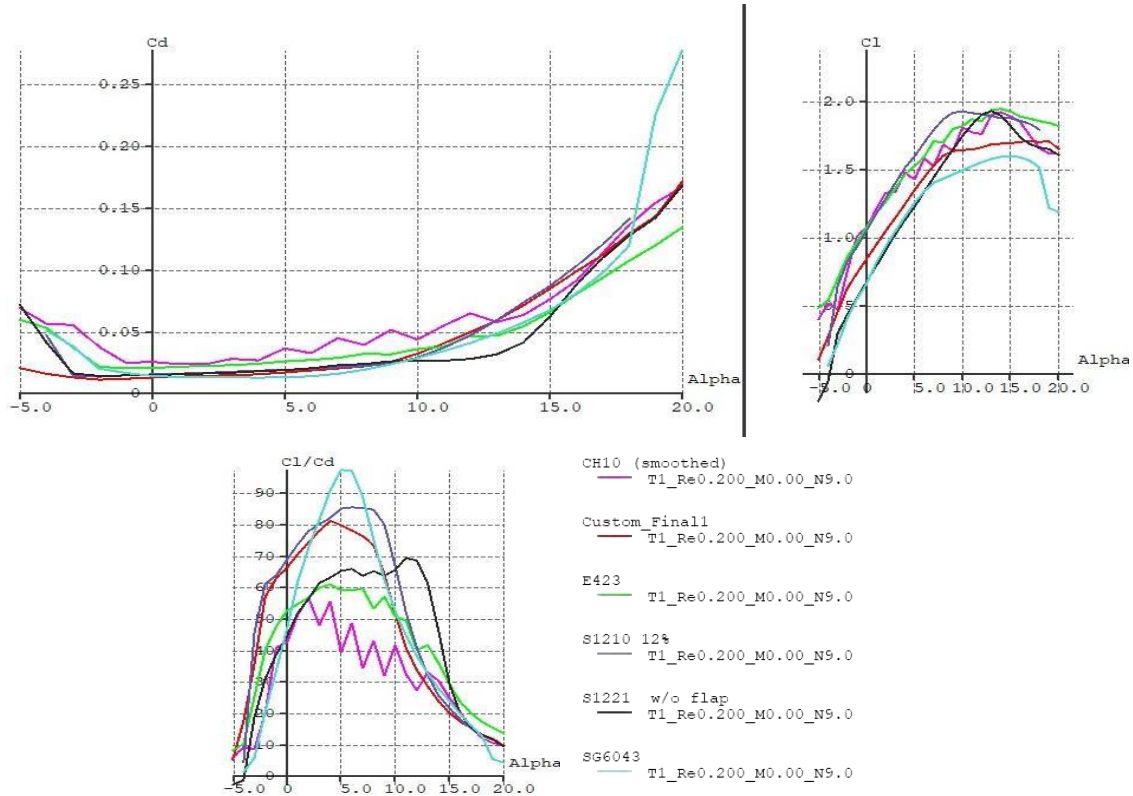


Figure 9: Analysis of Airfoil Performance Metrics at $Re=200,000$

The above process ensures that the final airfoil design meets the necessary performance criteria, paving the way for an optimized aerodynamic profile in our aircraft design and therefore, based on the XFLR analysis, S1221 airfoil was chosen. This choice was made based on its high CLmax, low drag and high aerodynamic efficiency at small angles of attack.



Figure 10: S1221 airfoil

In our assessment, we found that for the above airfoil, the lift coefficient is in the range $[0.8, 1.8]$ for angle of attack ranging from 0 to 20. The drag coefficient is in the range $[0.02, 0.06]$ contributing to lift-to-drag ratio spanning in the range $[20, 70]$.

4.2. Wing Design

Once the airfoils were selected, we proceeded with the next step of the preliminary design, which is to propose different wing configurations based on a conceptual sketch while verifying the static stability of the model by means of three-dimensional analysis in XFLR5.

The dimensions of the wing are limited as it must fit in the transport box of fixed dimensions. Having considered that and conducting the parametric study of chord length, wingspan and airfoil to find the optimum sizing, the allowable wingspan for our design ranged between 2.5 meters and 3 meters, with a maximum permissible distance from the wing to the tail section of 0.8 meters.

The wing geometry specification, including chord length, offset, dihedral angle, twist angle, and airfoil type, were meticulously selected to optimize performance. The final wingspan was decided to be 2.5m.

PARAMETER	SYMBOL	VALUE	UNIT
Root chord	c_w	0.25	m
Wingspan	b_w	2.5	m
Wing surface	S_w	0.48	m ²
Taper Ratio	λ_w	0.40	-

PARAMETER	SYMBOL	VALUE	UNIT
Aspect Ratio	AR_w	12.99	-
Root-tip Sweep	$\Lambda_{c/4, w}$	-1.72	degree
Wing dihedral	Γ_w	2.5	degree
Wing Twist	θ_{tip}	2	degree

Table 3: Key Parameters of the Wing Design

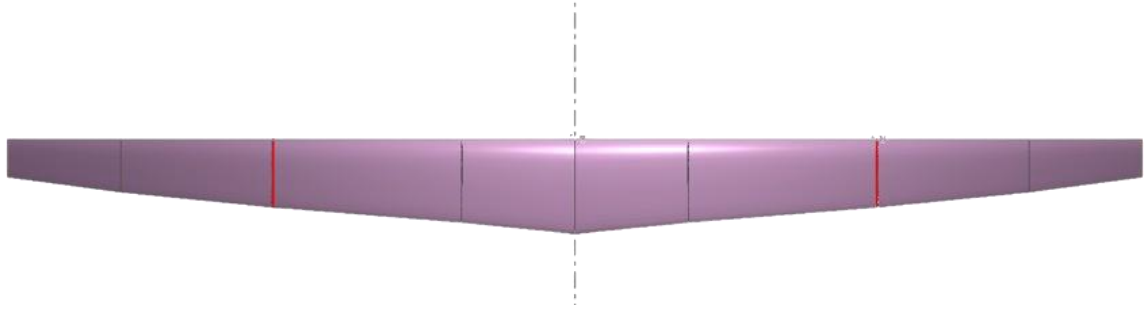


Figure 11: Initial Wing Geometry

The chord length decreases from 0.25m at the root to 0.10m at the tip. The dihedral angle was set at a constant value of 2.5° and the twist was maintained uniformly at 2.0° . The final design of the wing was adapted to facilitate easier manufacturing processes. These adjustments ensure that the aircraft can be produced efficiently without compromising on performance or safety standards.

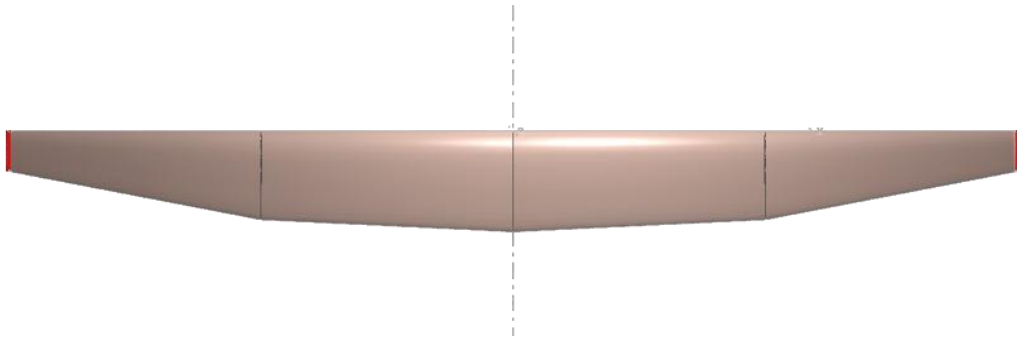


Figure 12: Final Wing Planform

4.3. Winglet Design

In our effort to optimize the aerodynamic efficiency of our aircraft, the design and selection of winglets played a crucial role. Various winglet designs were evaluated to determine the most effective configuration for our wing design.

The winglet design process involved creating detailed geometric models of various winglet types, including blended, sharklet, raked, and canted winglets. The geometries of the winglets

were developed using Autodesk Fusion 360. Each winglet type was tailored to fit the requirements of wing configuration. To assess the aerodynamic performance of these winglet designs, CFD simulations were conducted using StarCCM+. The primary objective of these simulations was to evaluate the impact of each winglet design on drag reduction, lift enhancement, and overall aerodynamic efficiency.

The CFD analysis yielded valuable insights into the aerodynamic performance of each winglet design. From the analysis, the sharklet emerged as the most balanced winglet for our wing design. Characterized by a sleek, upward-swept design, sharklets effectively reduce drag by smoothing airflow over the wingtips.

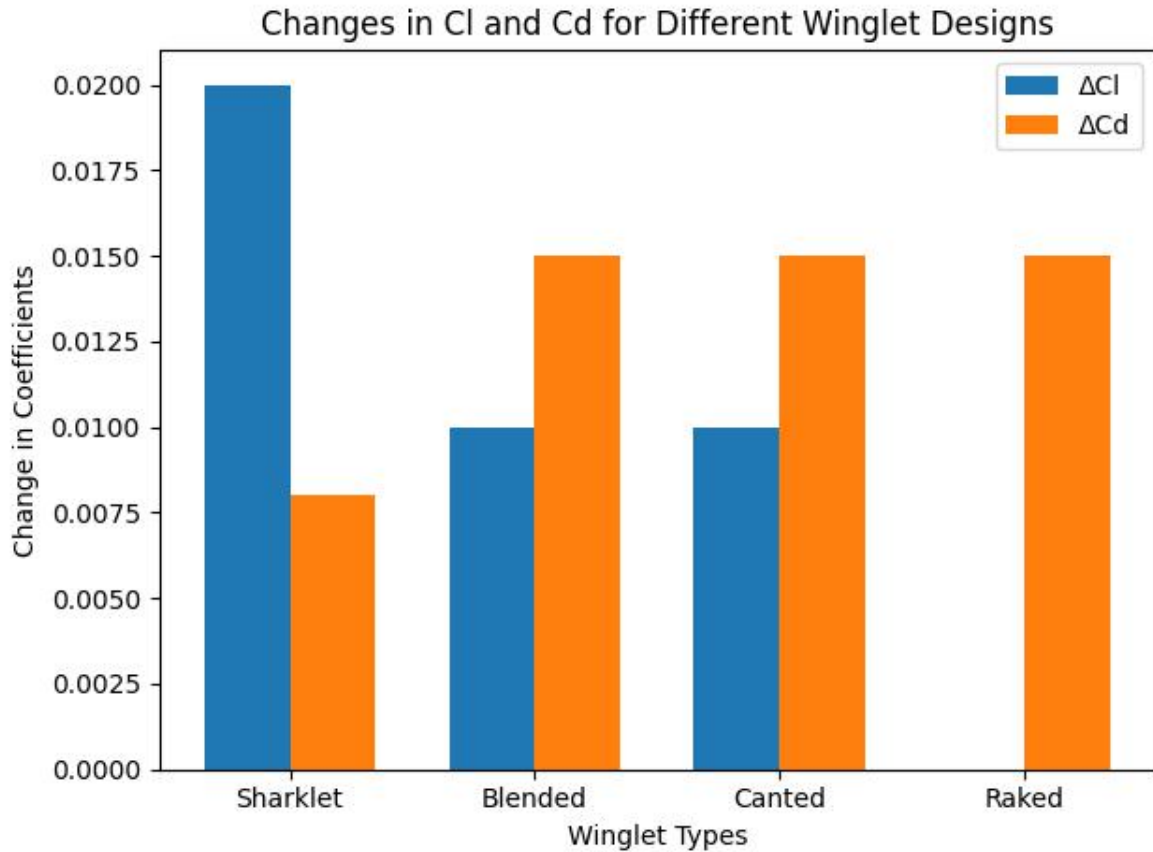


Figure 13: Comparative Analysis of Winglet Types

The above graph illustrates the changes in the lift coefficient (ΔC_l) and the drag coefficient (ΔC_d) for different winglet designs compared to a baseline configuration without any winglet. The baseline values are $C_l=0.65$ and $C_d=0.012$.

- The Sharklet winglet is the most efficient design, providing the highest increase in lift ($\Delta C_l = 0.02$) while maintaining a relatively moderate increase in drag ($\Delta C_d = 0.008$). This suggests that the Sharklet winglet significantly enhances aerodynamic performance.
- Both the Blended and Canted winglets offer moderate improvements in lift ($\Delta C_l = 0.01$) but come with a substantial drag penalty ($\Delta C_d = 0.015$). These designs are less efficient than the Sharklet.

- The Raked winglet does not improve lift at all ($\Delta Cl = 0$) and results in a significant increase in drag ($\Delta Cd = 0.015$). This makes the Raked winglet the least desirable option among those tested.

Therefore, the Sharklet winglet is the preferred choice due to a superior lift-to-drag ratio.

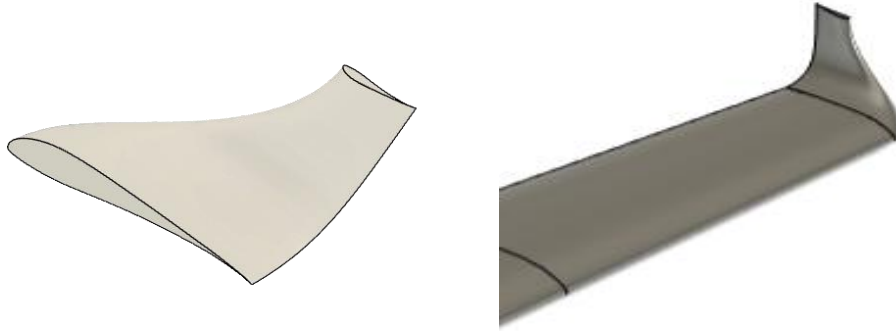


Figure 14: Sharklet winglet

4.4. Tail Design

The empennage, or tail assembly, of an aircraft is a crucial component for stability and control. For this project, a conventional empennage configuration was selected, comprising a horizontal stabilizer and a vertical stabilizer. This configuration is favored for its proven reliability and effectiveness in providing aerodynamic stability and control. This design ensures that the aircraft will perform reliably across a wide range of flight conditions, providing a solid foundation for further aerodynamic and performance optimizations.

4.4.1. Horizontal Stabilizer

Provides pitch stability and control. Positioned at the aft end of the fuselage, it includes an elevator for pitch control.

PARAMETER	SYMBOL	VALUE	UNIT
Mean aerodynamic chord	MAC	0.14	m
Elevator span	b_h	0.60	m
Elevator surface	S_h	0.08	m ²
Taper Ratio	λ_h	0.80	-
Aspect Ratio	AR_h	4.44	-

Table 4: Key Parameters of the Elevator Design

4.4.2. Vertical Stabilizer

Provides yaw stability and control. It is equipped with a rudder.

PARAMETER	SYMBOL	VALUE	UNIT
Mean aerodynamic chord	MAC	0.14	m
Rudder span	b_v	0.25	m
Rudder surface	S_v	0.03	m ²
Taper Ratio	λ_v	0.80	-
Aspect Ratio	AR_v	7.41	-

Table 5: Key Parameters of the Rudder Design



Figure 15: Empennage Design

Following the initial flight, it was determined that increasing the height of the vertical stabilizer would enhance pilot maneuverability, providing better control and stability.

A larger vertical stabilizer enhances the aircraft's ability to maintain directional stability, especially during low-speed operations. Additionally, it was observed that the actual hiatus between the control surfaces and the stabilizers was smaller than anticipated.

4.5. Aerodynamic analysis

To confirm all the calculations and simulations previously presented, a CFD simulation of the complete plane was performed in Star-CCM+. This simulation was performed at 0° of angle of attack, and at trim velocity.

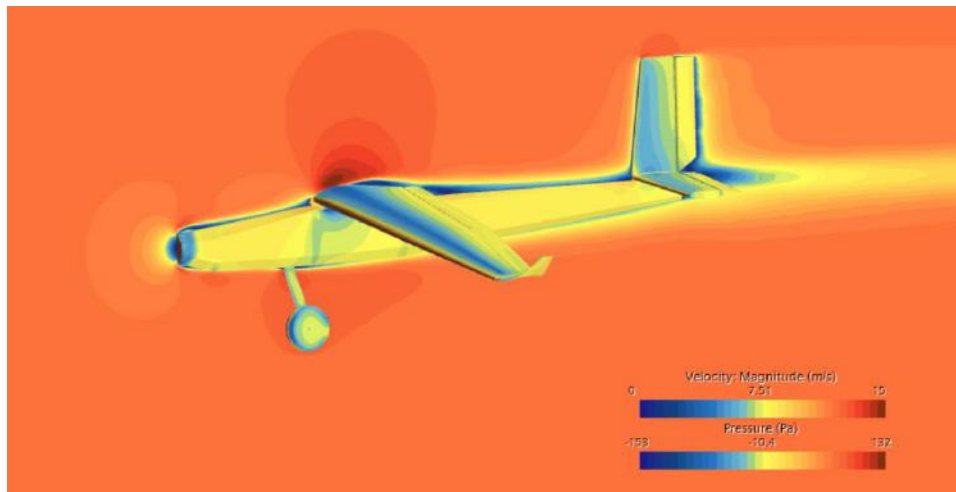


Figure 16: CFD study of the full aircraft

The distribution of pressure over the ECLifter is presented, as well as the velocity profile at the symmetry plane on Figure 16. The total drag coefficient obtained through this simulation was 0.032 and the lift coefficient was 0.62.

5. Stability and Control

5.1. Stability analysis

There are two different types of stability, static and dynamic stability. Statics is related to the generation of recuperative forces and moments, and dynamic stability indicates whether, under given reference conditions, the aircraft's response to any perturbations will reach a finite value.

In order to assume that an aircraft is statically stable, some requirements need to be fulfilled.

- The angle of attack at the null coefficient of momentum is not negative.
- The slope of the momentum coefficient with the angle of attack graph (C_{Ma}) has to be negative.
- The slope of the rolling momentum coefficient with the side slip angle (C_{Lb}) must be negative (lateral static stability criteria).
- The slope of the yawing momentum coefficient with the side slip angle (C_{Nb}) has to be positive (criteria of directional static stability).

Those values will be calculated in XFLR5 by a stability analysis 5 without input conditions. In Table 6, the values obtained by the ECLifter are presented.

	C_{Ma}=0	C_{Ma}	C_{Lb}	C_{Nb}
ECLifter	0.01	-1.146	-0.024	0.068

Table 6: Stability derivatives

The mass distribution of the ECLifter was essential in order to ensure a position of the center of gravity that would help the stability of the plane and reduce difficulties in maneuvering. The total operational empty weight of the plane is 2.475kg and is divided in the following elements:

Element	Fuselage and electronics	Main Wing	Horizontal Stabilizer	Vertical Stabilizer
Mass (kg)	1.32	0.950	0.135	0.07

Table 7: Mass distribution

The fuselage with the electronics inside of it will have its center of gravity 0.044 m behind the leading edge of the plane, and 0.6 below it. That will position the center of gravity of the ECLifter 0.076m before the leading edge and 0.018 below it (OEW configuration).

It is extremely important that the center of gravity is situated before the neutral point. In the case of the ECLifter, the neutral point is situated to 11.4cm of the leading edge, while the gravity center is at 7.6cm of the leading edge. The static margin is calculated using the mean aerodynamic chord.

$$S.M. = \frac{|x_{NP} - x_{CG}|}{MAC} = \frac{|0.114 - 0.076|}{0.250} = 14\%$$

Figure 17 represents the static longitudinal stability of the plane as the evolution of the coefficient of momentum with the angle of attack.

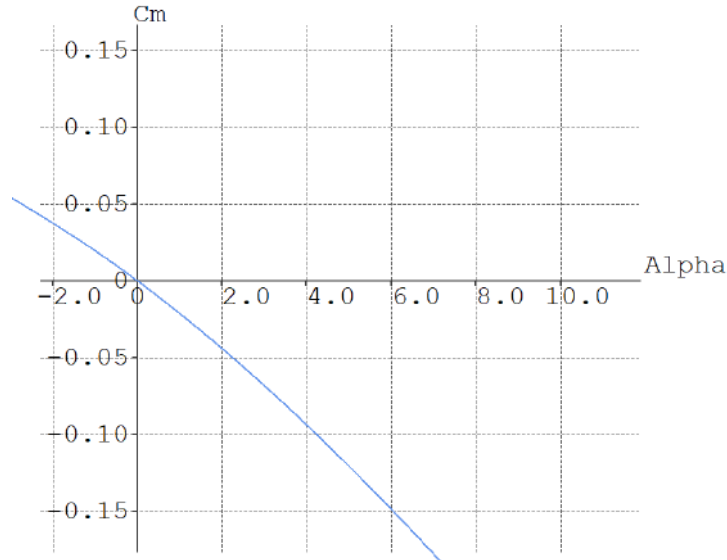


Figure 17: Longitudinal stability - Cm vs alpha

The dynamic stability is defined by the following graphics representing the stability eigenvalues or poles. These poles are considered stable when their real part is negative and they explain the answer of the aircraft when any input is introduced, and it can be calculated with XFRL5. It must be considered that some parts of the airplane, like the fuselage, were not introduced in the simulation and the real result may differ.

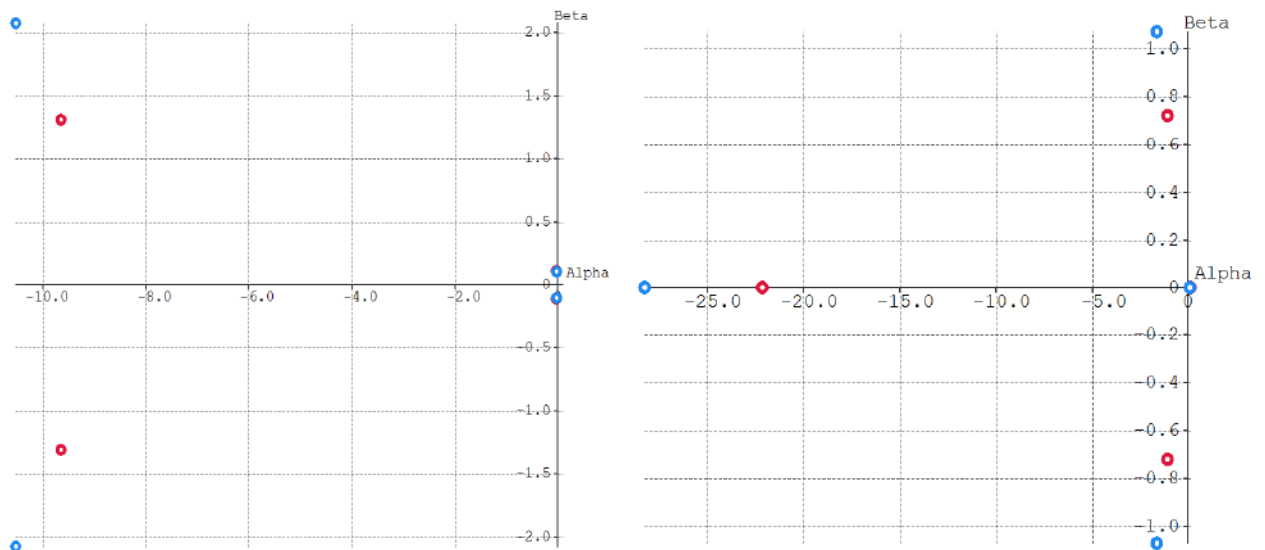


Figure 18: Dynamic stability root locus

In Figure 18, the poles of the final design with operational empty weight (OEW, red) and with maximum take-off weight (MTOW, blue), are presented. The first graph represents the longitudinal stability, which is in fact stable. In the lateral stability (second graph) it can be appreciated that one of the poles has a positive real value. This pole defines the spiral mode of the plane. At the beginning of the design process, this value was extremely big, and with an iterative approach was reduced significantly, its new value being low enough for a pilot to be able to easily counteract it.

After the second flight of the ECLifter, it was concluded that the spiral mode was still too big, so the center of gravity of the plane was moved forward in a successful attempt at improving the stability.

The MTOW configuration results in an increase of the real part of the poles in absolute value which shows the aggressiveness of the ECLifter increases with its payload while still being stable. Finally, this analysis provides also the trim velocity at OEW (48.6 km/h) and at MTOW (62.5 km/h).

5.2. Control surfaces

In the case of the ECLifter, there will be 3 movable control surfaces: ailerons, vertical stabilizer (rudder) and horizontal stabilizer (elevator). The dimensioning of the control surfaces can be performed using XFLR5 by defining those surfaces on the chosen plane and subjecting it to excitation.

By recurring to bibliography and using data from similar aircrafts, the following dimensions were chosen for those surfaces:

- Ailerons: will occupy 50% of the wingspan, and 20% of the chord
- Rudder: will occupy all the span of the surface, and 45% of the chord
- Elevator: will occupy all the span of the surface, and 30% of the chord

And as with the static stability analysis, some aerodynamic derivatives need to fulfill some control criteria to ensure the moving parts' dimensions are enough for the chosen aircraft.

- The slope of the momentum coefficient with the deflection of the elevator ($CM_{\delta e}$) has to be negative (quantifies the efficiency of the elevator and the climbing of the plane).
- The slope of the rolling momentum coefficient with the deflection of the ailerons ($CL_{\delta a}$) has to be positive (quantifies the efficiency of the ailerons).
- The slope of the yawing momentum coefficient with the deflection of the ailerons ($CN_{\delta a}$) has to be close to zero (corresponds to the yaw induced by the movement of ailerons).

Those values can be obtained by performing stability analysis in XFLR5 while changing the input conditions. In Table 8, the values for ECLifter are presented, as well as FAVARA, the plane representing the team XTRA2 in the Air Cargo Challenge 2022. This plane is used as a reference because of its similar goal (simple design, yet scoring many points).

	$CM_{\delta e}$	$CL_{\delta a}$	$CN_{\delta a}$	$CL_{\delta R}$	$CN_{\delta R}$
Definition	Elevator efficiency	Ailerons efficiency	Induced yaw	Induced roll	Rudder efficiency
Criteria	<0	>0	≈ 0	>0	<0
FAVARA	-1.7415	+0.3621	+0.0055	+0.0102	-0.0652
ECLifter	-0.7642	+0.3446	+0.0034	+0.0053	-0.0534

Table 8: Control Derivatives

The calculations result in similar values for both aircraft, except for the efficiency of the elevator. That will be translated into needing more deflection to achieve the desired movement, but it was concluded that this will, in fact, be positive due to the lack of experience of the pilot.

6. Structural Design

6.1. Introduction

Upon completion of the first prototype, one of the biggest problems encountered was the aircrafts manufacturing time and structural rigidity. The complex curving of the initial airframe proved difficult in keeping on design while joining pieces. Also, the low torsional rigidity of the tail-fuselage connection left much to be desired. From these lessons, a new mentality of simplification dominates the structural design of the aircraft. This results in the decrease of former and stringer numbers while also increasing their thickness. This re-design allows for much greater strength, accessibility, and rigidity when compared to the prototype.

Similar to before, the structural development continued with careful attention paid to the rules. Desire to decrease the loading time led to increased size and numbers of hatches. Also, specialized holding cells are designed for the loader to easily access the cargo. Furthermore, the avionics bay is extended to be the full length of the noise section due to the improved materials acquired.

6.2. Fuselage Design

The fuselage of the ECLifter has been separated into two sections for convenience and reliability. Primarily, the desire to transport mass and volume required a strong and spacious aircraft. The 750mm max length restriction and the need for a hiatus of at least 500mm encroached the split fuselage. The following two subsections discuss the sections of the fuselage. For the main body, the following is discussed: avionics, ball bays and loading methods, wing – fuselage connections, servo placements, landing gear, and hatches. The latter touches on the tail body and its tail– fuselage connection, loading scheme, servo connections and landing gear.

6.2.1. Main Body

The main fuselage section of the ECLifter houses all the critical components of the aircraft. It mostly consists of 4 stringers, 6 formers, and 4 bulkheads. All components are made of a 5mm plywood and attached to this are various other subcomponents. This can be seen well in Figure 19, where an exploded view of the main body is given.

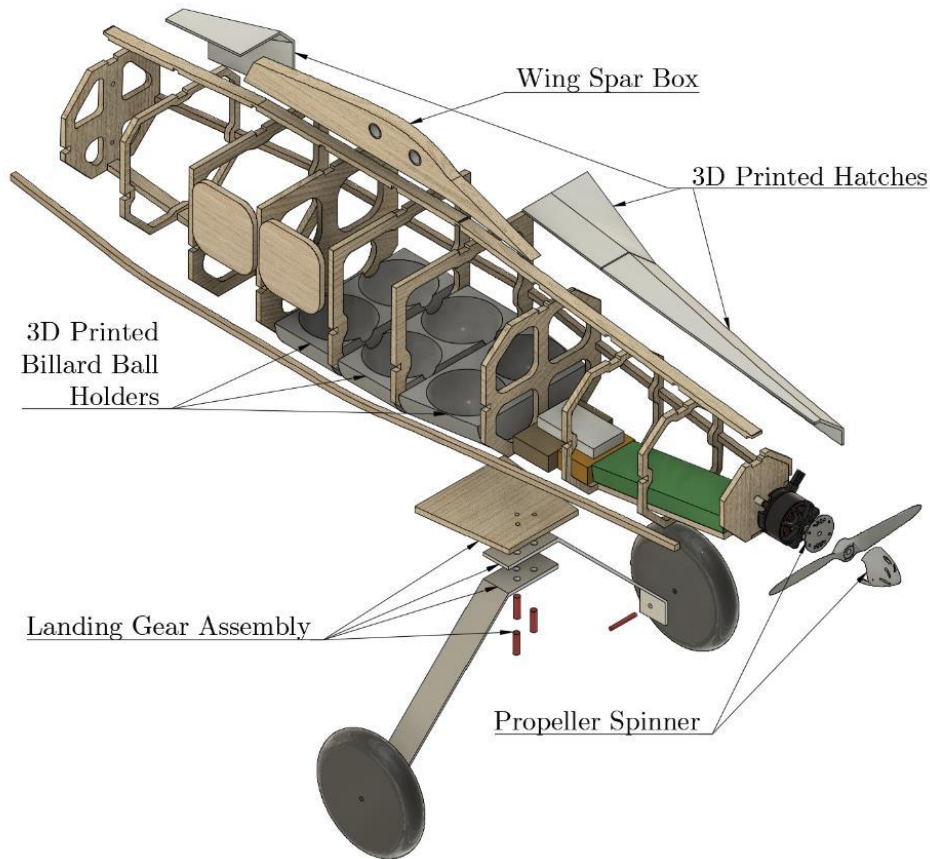


Figure 19: An exploded view of the main body fuselage.

Various sub-assemblies latch onto the main frame, including landing gear, wing – fuselage joints, payload storage, and hatch access.

Starting at the nose of the airplane, the body is sized in such a way that we do not need to remove the motor for loading into the box. The motor holes are lasered into the firewall to ensure accuracy of the thrust line. After difficulties with small access hatches on the prototype, over a quarter of skin is removed from nose section to make way for a large avionics hatch. The weight of the 3D printed hatch is kept to a minimum thanks to special lightweight PLA. Also, a shelf and straps are installed to securely load the powerplant.

Past the nose, the billiard balls are housed directly under the wing. This is done intentionally, to minimize the effects of altering payload configuration on the center of gravity. The billiards are kept contained with 3D printed “egg cartons”, which are formed to sit in junction with the lower skin and stringer. The idea is to load the 6 billiards in the holders and stack more atop while using added rubber bands to secure the additional balls. Two hatches access the Billiard section. One, a large 3D printed hatch towards the front, and the other a

smaller wooded hatch at the back. Both will be used during loading but for different purposes. The larger will see a hand helping to arrange the balls. The other is purely a loading zone.

Below the billiard section is the load path for the landing gear. A large 5mm plywood reinforcement is added to the belly for strength during landings. On the maiden flight of the aircraft, the only component to suffer damage was the landing gear. Indeed, the small connection area of the prototype was shattered after landing too hard. The wheel assembly itself is off the shelf. The holes of the landing gear strut are lasered into the reinforcement plate to ensure level wheel.

The joint that allows for the wing connection sits on top of the body. It is a simple 10mm plywood joint that is shaped to match the root airfoil. The holes of the connector are fitted with PVC tubes to ensure proper alignment and easier assembly. The connector is attached directly to the top stringer to ensure a strong connection for the load path.

Finally, the rear of the main body is dedicated to ping pong storage. Like the billiard section, two hatches are situated across one another for the access of the payload. No special housing is made for the ping pong balls. This is because once you stuff enough balls into the cavity and tape it shut, there will be no moving balls. Without the presence of ping pong ball holders, loading and unloading can proceed much quicker.

Overall, the integrity of the main body fuselage is fortified by its 5mm balsa skin over the bottom of the aircraft. This thick and light belly will absorb strong blows and in the tragic happening of an accident, it will be relatively easy to fix.

6.2.2. Tail Body

The tail body fuselage has many less tasks than its predecessor. Most importantly, it must have high torsional rigidity to allow for smooth empennage operations. The torsional stiffness of the prototypes tail connection was extremely low, leading to the adaptation of a stronger materials. These materials allowed for less formers and stringers than our prototypes architecture while still massively increasing the rigidity. Similar and smaller to the latter, the tail body consists of three bulkheads, two formers, and four stringers. An exploded view of the tail body is given in Figure 20.

Unlike the 5mm plywood used in the main body, the tail body used a 4mm ply instead as the structure isn't as heavily loaded. The leading bulkhead has screwed holes that match up with the end bulkhead of the main body. More on this in the manufacturing section. Flush with the front bulkhead also sit the elevator and rudder servos. The servos are not shown in the CAD as their precise placement is difficult to estimate. The rudder is connected in pair with the rear landing gear to increase taxi handling. The tail landing gear is a simple off the shelf component that is strung together using fishline. It is a great, low weight, effective component.

To connect the empennage to the body, a 3D printed component is designed. It has four holes in it, two vertically and two horizontally. The verticals are stopped by the top stringer which the connector joint sits on. Whereas the horizontals pass through the to join with the other half. Various iterations of the component were seen, a lightweight PLA was found to be reliable while remaining cheap and fast to produce for its small size.

There is only one hatch in this section; it is responsible for servo maintenance, screw access, and ping-pong ball loading. Indeed, thanks to the architecture, the plane can fit up to 20 ping pong balls in its tail section. Again, there are no dedicated housing spots for the balls to populate. This is done with the desire for swift loading and unloading to transpire. In Figure 20, an exploded view of the tail is given.

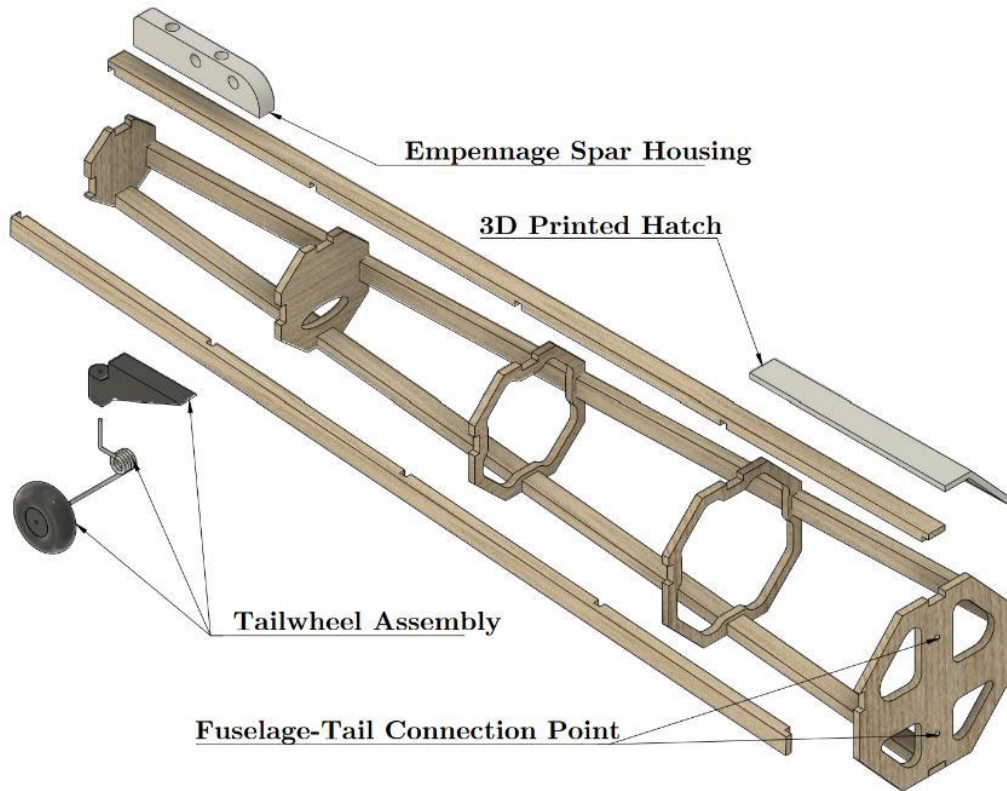


Figure 20: An exploded view of the tail body fuselage.

6.3. Wing Design

Like the fuselage, the wing is split into two sections to fit in the enforced box. An inner panel and outer panel are designed to work in concert to create a 2.5 m wingspan. This is done through classical architecture of spars, ribs, and doublers. The inboard and outboard sections are very similar, save for their spar numbers. Thermoforming plastic wrap is chosen for the skin due to its light weight, low cost, and reparability.

6.3.1. Inner Wing Design

The inner wing has four 8mm aluminum spar-tube connections running through it. One set goes across the fuselage spar connector and into the opposite side wing. The flap runs the whole length of the span, therefore for evenly applied torque the servo actuates from the center of the span. The other set runs into the outboard wing section. In Figure 21, an exploded view of the inboard wing is shown.

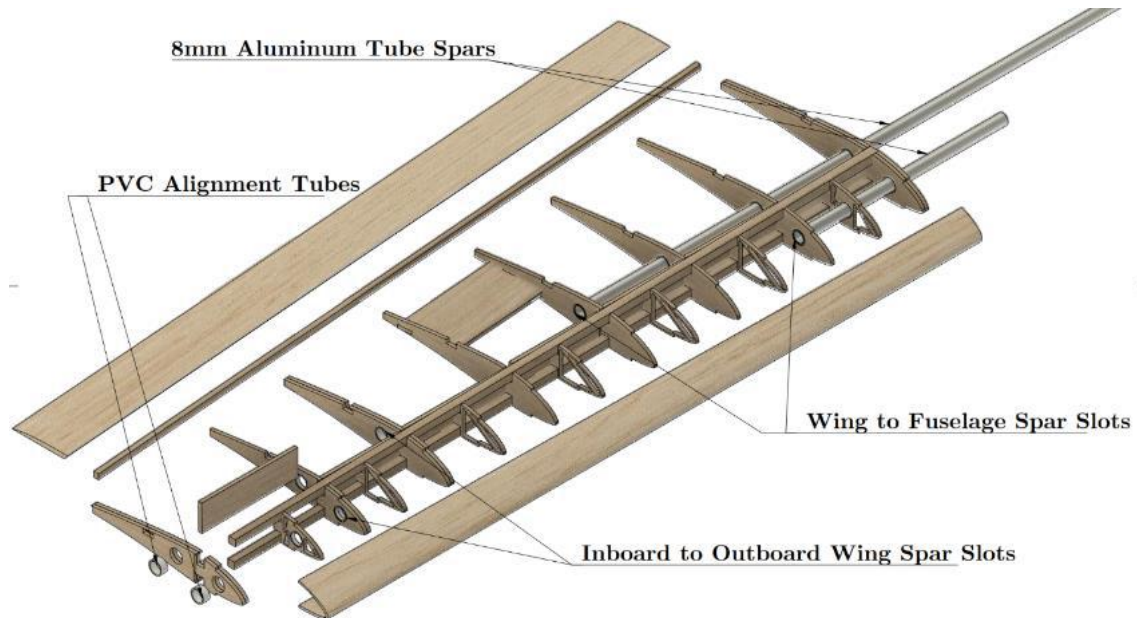


Figure 21: An exploded view of the inboard wing.

Like the spar holder of the main fuselage, there are PVC alignment tubes embedded into the wing to allow for easy spar installation. For the flaps, control surface pads are installed at the midspan to ensure the torque is distributed evenly.

6.3.2. Outer Wing Design

As previously stated, the outboard architecture of the outer wing is identical to the inner wing. The leading edge is made of a 3mm balsa, the control surface is made from 5mm balsa, and the 8mm aluminum spar-tubes run about a quarter into the structure. Most of the wing structure uses 3mm plywood (ribs, doublers, square spars).

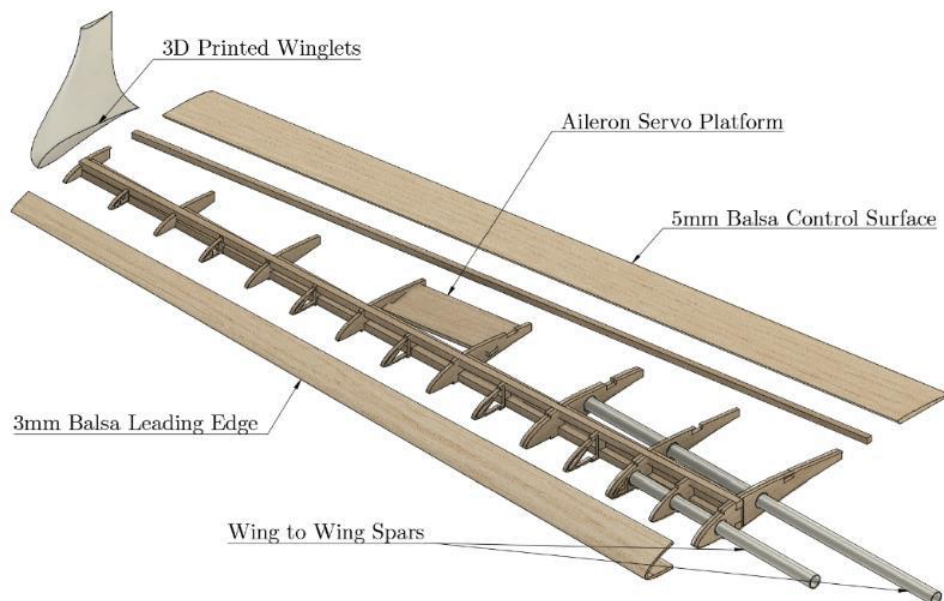


Figure 22: An exploded view of the outboard wing.

What is newer are the winglets installed onto the tips of the outer wings. The winglets are made of the lightweight PLA material. While they have a different airfoil than that of the main wing itself. They're large enough to match the face with that of the tip. In Figure 22 an exploded view of the outboard wing is given. The 3D printed winglet offers a flat surface for which joining is made simple.

6.4. Empennage Design

The empennage is completely synonymous with the wing section. The only major change is the use of 6mm aluminum tube-spars, this can be seen in Figure 23.

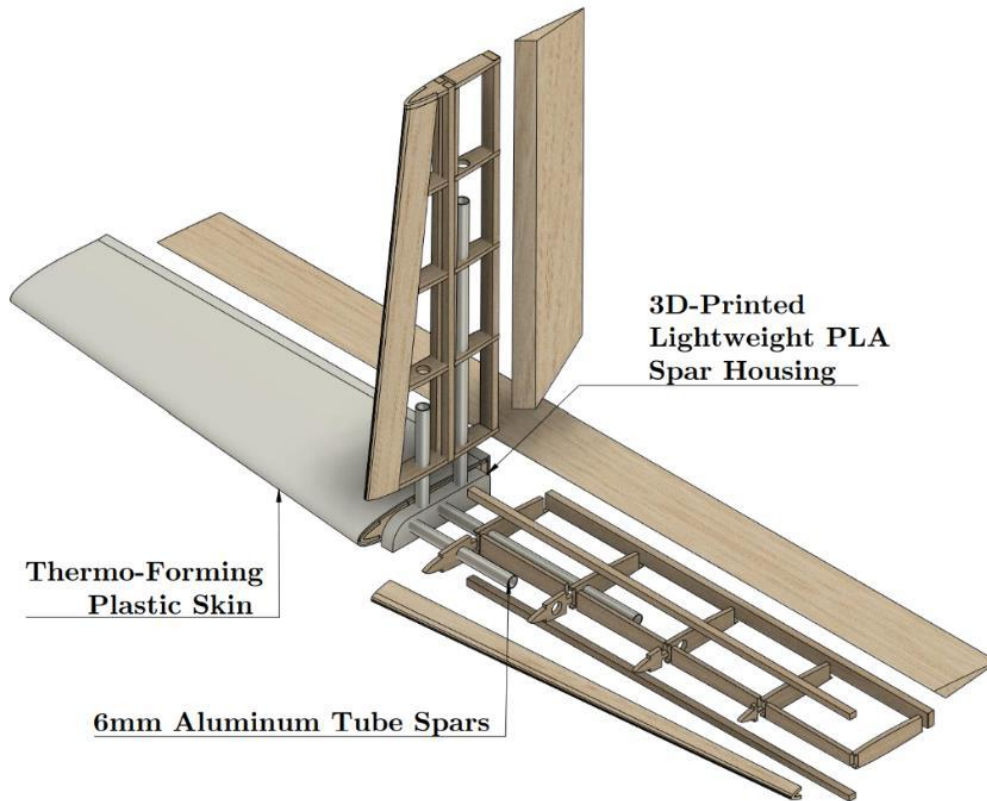


Figure 23: Exploded Views of the Rudder and Elevator

6.5. Center of Mass vs Payload

To characterize the effect of billiard ball placement on mass center, Figure 24, is produced using Fusion's material editor among other tools. To start, all components of the aircraft are assigned with custom materials that represent the true density of materials used in the aircraft. These true values came from measurement of stock materials weights and calculations of volumes. Once all materials are assigned, and the assembly is made, it is a simple click to calculate the center of mass and relate it to the leading edge of the wing.

The sequence starts with a fully loaded bay of eight balls. Six rest in their holders and two in the middle pockets made by the others. Six balls therefore represent all holders s loaded. To remove the remaining six, the method is from front to back. So that four balls represent

the two back rows of holders be occupied. And lastly, two balls being placed in the holders furthest aft of the aircraft.

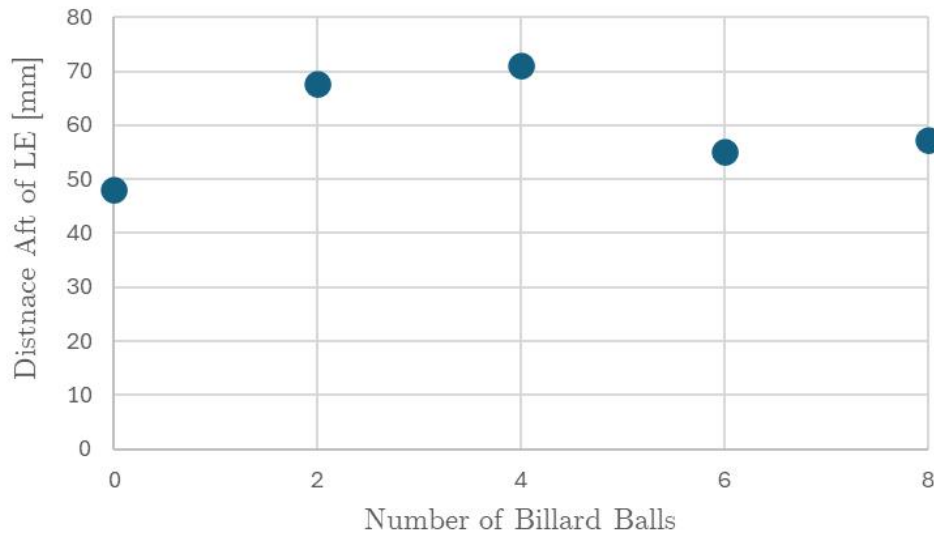


Figure 24: Distance to the C.G. from the L.E. of the wing as a function of billiard balls.

Our desired range was to be between 50 and 80mm, the configuration seems to be well within the limits. Furthermore, there appears to be an inflection point at from 4 balls leading to a decrease in distance. The balls closest to the front are taken first, then the ones behind

6.6. Structure Simulation

To validate our design, 2 simulations were performed using COMSOL: static analysis of wings and landing gear. In both cases, load conditions are the mass of full plane. Results of the landing gear simulation. simulations are presented in Figure 25.

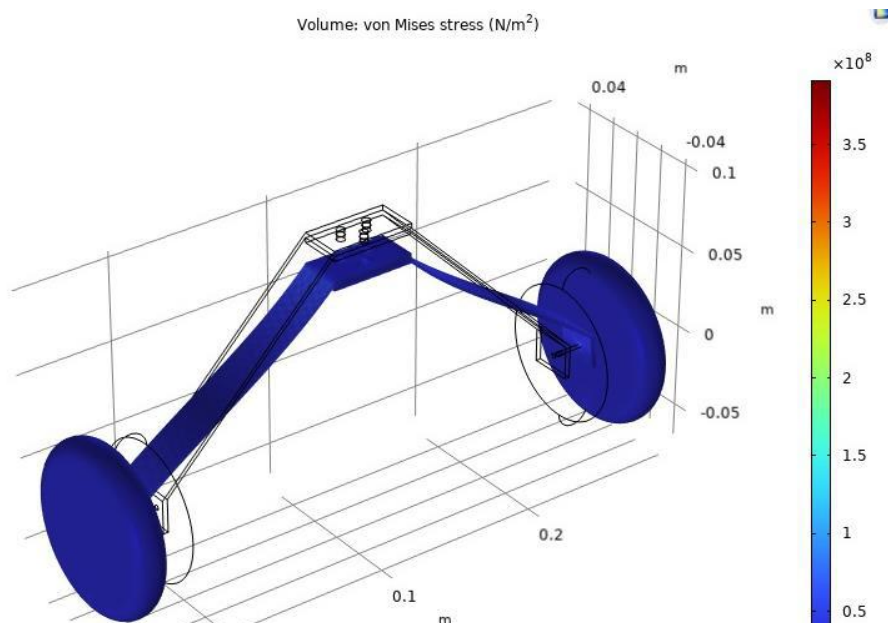


Figure 25: Landing gear under static loading in COMSOL

For the landing gear, the critical point is the screw joint of the wheel, and the steel connector, with a maximum stress of 440 MPa. For the wing, the critical point is the aluminum half wing rod joiner, with a maximum stress of 300 MPa can be seen in Figure .

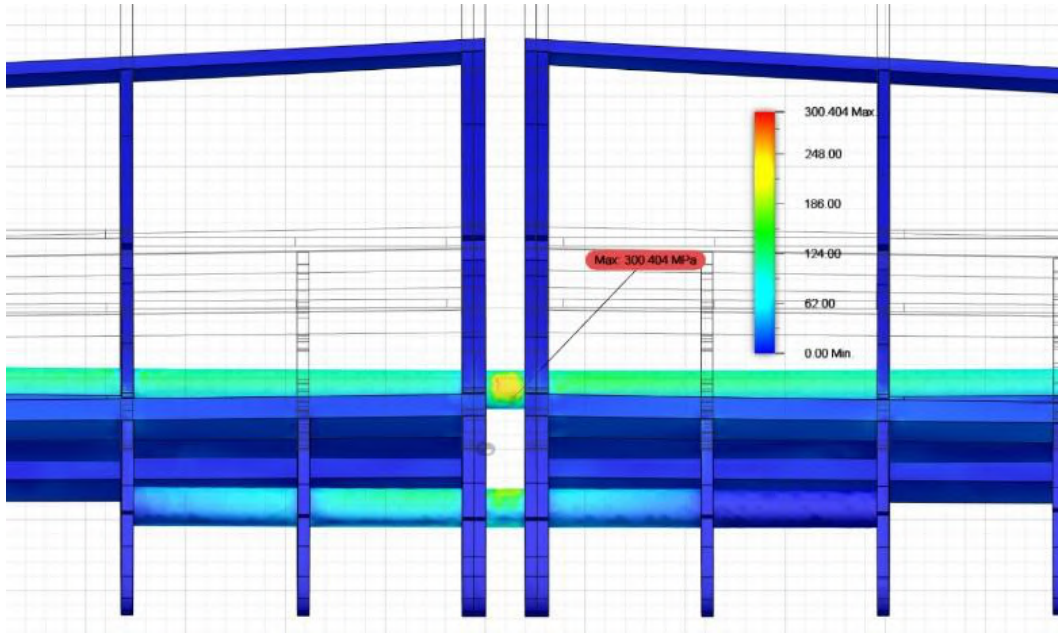


Figure 26: Top view of the wing under static loading in COMSOL

6.7. Box Design

To compete in the XtraChallenge, a box is needed for the aircraft to be loaded and unloaded from. The choice is made to construct the box from wood for many reasons. Firstly, it is relatively cheap, strong, and easy to manufacture. Also, wood can be easily altered for installing upgrades or storage rails. In Figure 27, the ECLift Box is shown.



Figure 27: ECLifter's Box, made of wood

7. Manufacturing

This section will detail methods used in the manufacturing of the ECLifter. Located at École Centrale, the FabLab facilities were exclusively used. 3D-printing, laser cutting, soldering, and many other processes were carried out in the facility. Best practices were slowly developed through the manufacturing process. As the relative experience among members with construction is low, quality parts were made thanks to patience and optimism.

The primary materials used in the creation of the ECLifter was wood (balsa, ply, MDF), lightweight PLA, adhesives (CA, wood glue, hot glue), and various other plastics. The most used material was the 5mm plywood. This strong and tough wood constitutes the backbone of the aircraft. While wood glue was used extensively on the prototype aircraft, almost all connections are made with CA on the final assembly for the added strength and decreased cure time.

7.1. Preparation

Before construction can begin, many steps must be taken beforehand. Most namely, laser cutting and sometimes 3D printing. Once designs were complete, .dxf files were generated and developed using Inkscape software. Always trying to maximize the use of our parts, Figure 28 shows the .dxf file for the fuselage where comment's edges touch to decrease the required number of laser passes. Even with the touching edges, it will take the laser over 40 minutes to cut through the tough 5mm plywood. The parts are also labeled for easy assembly later. For some materials, the laser needed to be tuned, leading to better finished parts with less char.

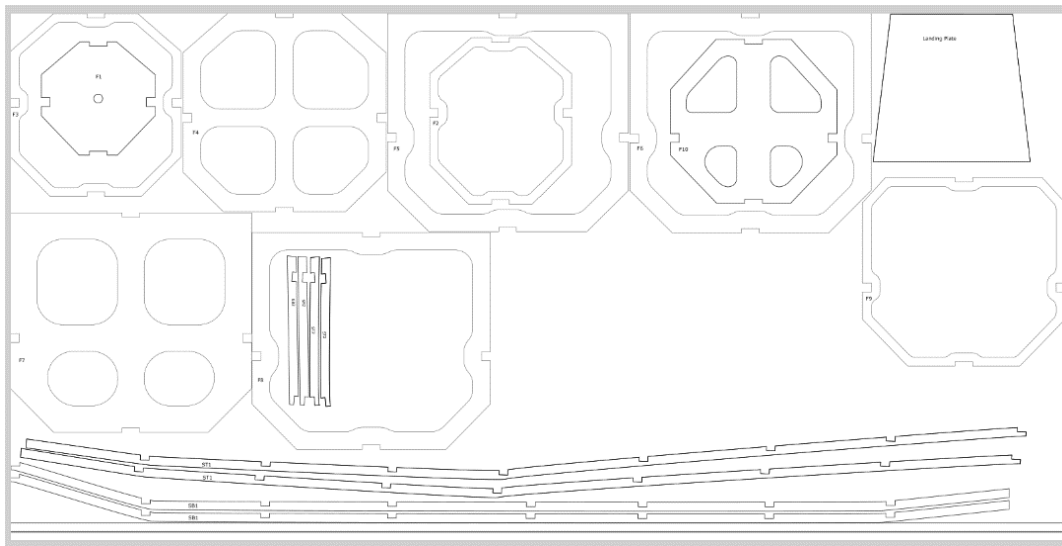


Figure 28: The .DXF file of the main body fuselage

The big downside of using wood with lasers is the char that is left on the wood's surface after cutting. Not removing the charred surface before applying adhesives would greatly reduce the joints strength. Therefore, care was taken to always lightly sand away the char from any areas that would see glues. Of course, great care was also taken to not over-sand the part leading to a loose fit.

7.2. Fuselage Assembly

Once all parts were assembled, steps were taken to precisely and efficiently construct the fuselage. The steps were very tedious on the prototype aircraft. With its complex curving, it was extremely difficult to ensure the stringers were aligned correctly. The need for a simpler geometry thus became abruptly clear, leading to fuselage revisions in the 2nd aircraft. Figure 29 gives a look at the prototypes internal structure, and all its curves.



Figure 29: The internal structure of the first prototype ECLifter

Poor torsional stiffness led to the addition of cross-bracing members between formers. While this helped the torsion, it did not help the skinning process. For the second aircraft, a much smoother process was developed. The refined geometry of the spars and formers led to parts fitting together very easily. So, starting with the middle three formers, the top stringer was attached with great care to ensure their respective 90-degree angles in both pitch and yaw. The middle three formers share almost identical height allowing them to a great support base for the other, more original, pieces. Once all formers are attached to the top spar, the bottom stringer followed by the sides are attached. The following process is carried out for the tail body fuselage.

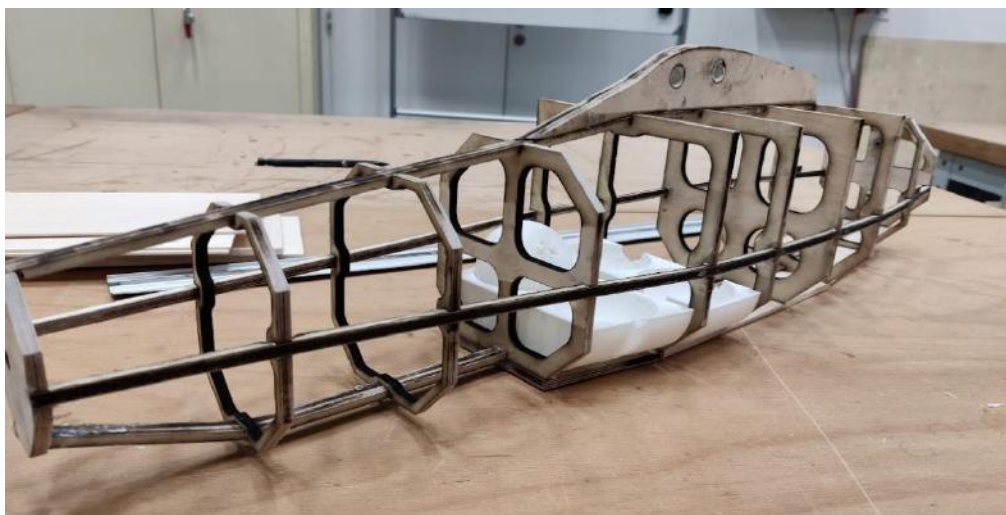


Figure 30: Manufacturing of the revised fuselage structure

With the formers and stringers in place, modifications are added before the skin is applied. Namely, spar joints are glued to their respective top spar using superglue the landing plate is installed, and tail servos are added. For skinning the lower half, the belly, of the main fuselage, 5mm balsa is used to offer more impact resistance on landings. For the upper half of the main fuselage, and for all the tail body fuselage, 4mm balsa is used for its flexibility and ease of sanding. Also in the upper half, specific sections are not skinned to allow for 3D printed hatches to fill the place.

The hatches are secured in place with magnets and glue. Grate care is taken to ensure the magnet has strong support both on the hatch and in the fuselage. For the fuselage's magnets, recessed wooden corners are built with balsa to protect against fast moving hands.

The tail body to main body connection, landing gear, and motor mount all consist of simple screws and holes. They are all installed with a simple twist, thanks to the lasered holes instillation is simple and fast. Also, the 3D printed PLA billiard ball holders are glued into place, taking support from both the bottom stringer and the skin. Lastly, a lightweight filler material is applied to cracks. Sandpaper is then applied, and the cycle repeated until the desired smoothness is achieved.

7.3. Wing Assembly

Like the fuselage, once all parts were assembled, steps were taken to precisely and efficiently construct the wing. With the main wings there were considerably more parts to keep track of. Yet despite this, the prototype wing was a huge success. Most impressively, the leading edge of the wing was fitted with balsa by first wetting it and tacking it in place with wire. In Figure 31, the prototype wing is being fitted with its balsa leading edge.

To begin the wing construction, the wooden ribs with spar holes are slid over a PVC tube to ensure their straightness. Then, while still on the PVC tube, the wooden top and bottom square spars are glued. So that, when the PVC tube is removed and the other ribs are installed, the straightness of the guiding square spars is insured.

Once the square spars are applied, the spacers are placed in-between the ribs to act as a shear web. The trailing edge is then attached, along with the PVC fittings around the spar holes. Finally, the servo mounts are installed along with the servos themselves. The wing now has all structural members joined together, save for the winglet.

The wing is then ready to be skinned, this is done using a hot iron and thermoforming plastic. Great care is taken to create taut and flush skin. This process is quite tedious and not for the impatient. When the skin is applied, the control surfaces can finally be attached.

The control surfaces are all made from 5mm balsa, sanded down on one face to provide a chamber and whittle on the other to provide room for deflection. Simple tape is used to connect the control surface to its host. Finally, with the servo linkages attached, the wing is ready to be used. The exact same processes go for the empennage, except for the servo placements which happen in the tail fuselage and run via linkages to the rudder and elevator.

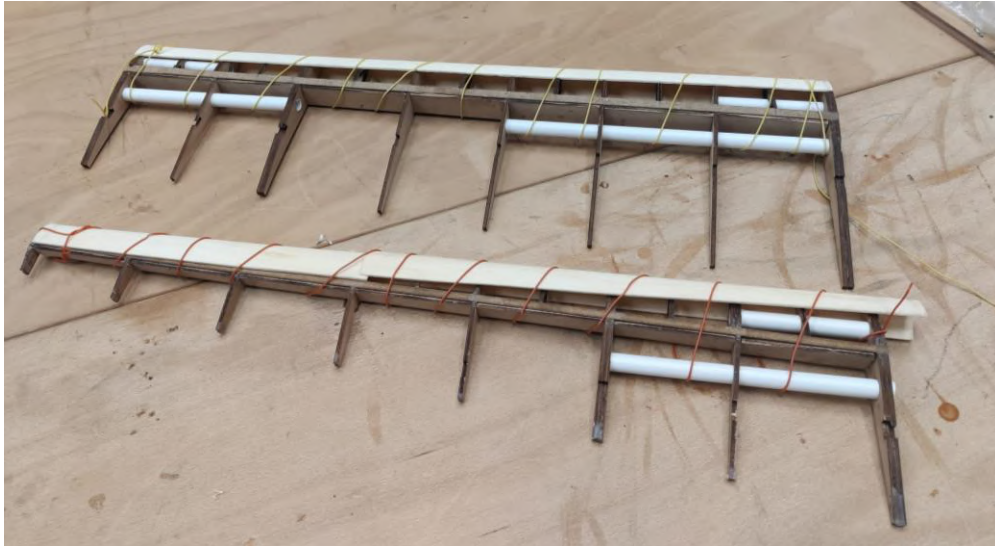


Figure 31: The 3mm balsa is being fitted to the leading edge of the wing. On top is the inboard section, on bottom the outboard. The process involves soaking the wood in water and allowing it to dry while deformed, retaining its formed shape

7.4. Assembled Aircraft

Finally, the aircraft is ready to be assembled. A rendered version of this is given in the following figure. It is interesting to note that up to 14 billiard balls could fit in the bay, and over 60 ping pong balls could fit in the rear two bays. While the MTOW required for this is too great for our aircraft, it is exciting to think that with a propulsion upgrade, the lifter could carry much more cargo.



Figure 32: Render of the assembled ECLifter

8. Aircraft Performance

To sum up, we present a compilation of the most important values used to assess the aircraft's performance. Analyzing our final model in Xflr5, we obtain:

PARAMETER	VALUE
ρ	1.18 kg/m ³
OEW	2.45 kg
MTOW (w/o flaps)	4.04919 kg
MTOW (with flaps)	4.63775 kg
C_{Lmax} (w/o flaps)	1.471
C_{Lmax} (with flaps)	1.829
C_D (w/o flaps)	0.0586 +0.0408 (-0.368 + C_L) ²
C_D (with flaps)	0.0776 +0.120 (-0.699 + C_L) ²

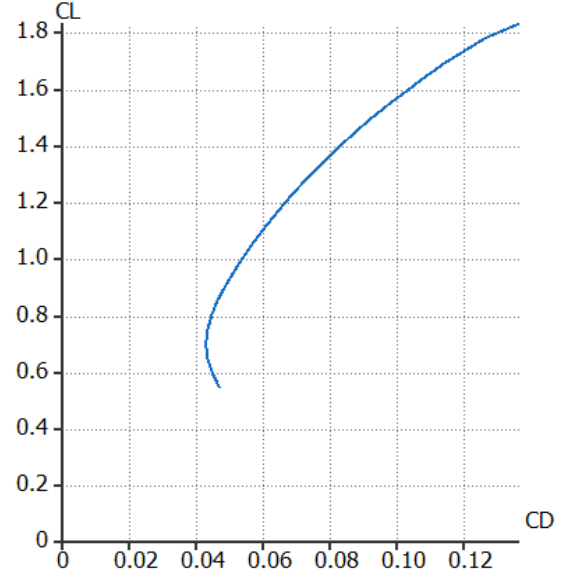


Figure 33: Aircraft Polar

Table 9: Performance parameters

We proceed to calculate the take-off velocity:

$$V_{TO} = 1.2 \times V_{stall} = 1.2 \times \sqrt{\frac{2W}{\rho S_w (0.8 C_{Lmax})}} = 43.1913 \text{ km/h}$$

With $W = m_T \times g$ with m_T total mass.

The take-off distance is given by the next equation

$$x_{TO} = m_T \int_0^{V_{TO}} \frac{V}{T(V) - \frac{1}{2} \rho V^2 S_w (C_D - \mu_d C_L) - \mu_d m_T g} dV$$

To deduce the payload mass equation, we simply use the definition of MTOW:

$$MTOW = MPL + OEW \rightarrow MPL = MTOW(\rho) - OEW$$

$$MPL = \frac{1}{g} \left(\frac{1}{2} \rho S_w V_{TO}^2 C_{Lmax} - OEW \right)$$

$$MPL = 3.322 \times \rho - 2.45$$

Using that last equation, we obtain the following figure:

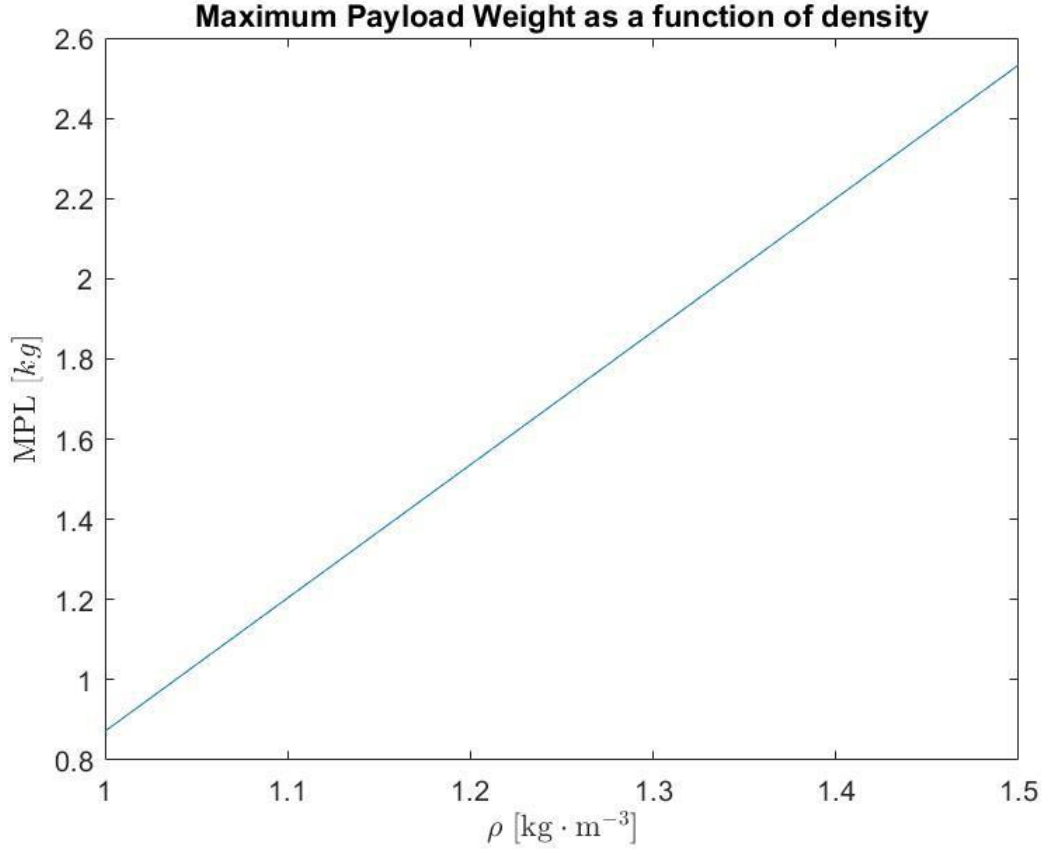


Figure 34: Payload Prediction as a function of air density

As for the trim, we can obtain the velocity and the flap deflection from the balancing equations

$$L = \frac{1}{2} \rho S_w V^2 C_L \approx W$$

$$V_{trim} = \sqrt{\frac{2W}{\rho S_w C_{Ltrim}}} = 66.8817 \text{ km/h}$$

$$D = \frac{1}{2} \rho S_w (C_{D0} - K(C_{Ltrim} - C_{Lmin,D})^2) V_{trim}^2 \approx T(\delta_{p,trim})$$

$$\delta_{p,trim} = \frac{\frac{1}{2} \rho S_w (C_{D0} - K(C_{Ltrim} - C_{Lmin,D})) V_{trim}^2}{(a - b(V_{trim} - c))}$$

9. Innovations

The ECLifter has a relatively traditional aerodynamic design and uses classical woodworking techniques for manufacturing. It features, nonetheless, some unique characteristics that make it a truly innovative aircraft prototype. Including:

- The use of **lightweight additive manufacturing materials** for key parts like hatches or non-structural elements like winglets and other accessories.
- The implementation of a **SLS chemistry based powerplant**, which optimizes the thrust by supplying a higher voltage to the motor without violating regulations. Improving the performance of an otherwise quite limited propulsion system.
- The overall **simplicity of the structure**, that has been designed in an intelligent way to allow quick and easy fixes during the competition without leading to penalties. Multiple sets of wings and spare parts fit together inside the same transportation box.
- The **low-cost, short-time** approach and production scalability as a commercial product that can be manufactured in series with less than 100 parts to be assembled. Compared to the +300 parts average of other planes with a classic wood structure.

10. Lessons learned

The development of the ECLifter has been an exhaustive iterative process and a great learning experience for the members of all departments. The full sequence of designing-building-flying this aircraft really gives a new perspective to all the theoretical knowledge that is taught to students in engineering schools.

The ECLifter started as a design with the intention of making the job easier for the pilot and to bet on safe choices rather than high-risk-high-reward alternatives. A stable aircraft with a conventional tail, high wing, and many off-the-shelf components.

However, it has still proven to be challenging in many ways. After a relatively successful first flight, it was decided to increase the wet surface of the tail fin based on the feedback from the pilot. In a second flight, some strange behaviors repeated with a catastrophic



Figure 35: Aftermath of the second flight of the ECLifter

result. The ECLifter crashed after a stall during the approach. The nose was shattered, and the wing broke in several parts.

A decision was made to shift the center of gravity towards the leading edge. It was also loaded with a portion of the Payload to increase its mass. The wing was repaired, and the fuselage was rebuilt. The trim angle of the elevator was adapted, and its control channel was configured with an exponential lever to prevent PIOs (Pilot-Induced-Oscillations).

All these improvements contributed to the success of the 3rd flight test.

11. Closure

The entire ECLift team is very proud of the results obtained during the few months dedicated to the project. For most of the members, it has been a first-ever hands-on experience; very few students had previous expertise in the field of aircraft design or even aeronautics.

The difficulties extended further from the technical challenges. Never before had a group of master's students tried to start an initiative of this kind at ECL. In the beginning, there was no mechanism for funding, no organization or structure designated to welcome and support such type of project.

But through close collaboration, and a strong willingness to learn and create a great community of highly passionate students with a common goal, all the adversities were overcome. An example that with the right people, the sky is literally the limit.



Figure 36: Final approach during a test flight on June 30th, 2024

We look forward to the XtraChallenge 2024, to learn from the best, and to share with the other participants more stories just like ours.

References

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- [4]. Prieto, M., Escarti-Guillem, M. S., & Hoyas, S. (2023). Aerodynamic optimization of a VTOL drone using winglets. Results in Engineering, 17.
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Annex I: Budget Breakdown

CONSTRUCTION MATERIALS	€/UNIT	UNITS	TOTAL
3 U. Balsa 50/10 100X10CM	10,30 €	2	20,60 €
5 U. Balsa 20/10 100X10CM	11,70 €	2	23,40 €
AVIATION PLYWOOD 6/10 30X60CM	14,50 €	1	14,50 €
AVIATION PLYWOOD 50/10 30X60CM	14,90 €	5	74,50 €
AVIATION PLYWOOD 40/10 30X60CM	11,90 €	2	23,80 €
ORACOVER 2M	28,50 €	2	57,00 €
ORACOVER FLUO 2M	33,30 €	2	66,60 €
WING JOINER Ø10X8MM	2,80 €	4	11,20 €
WING JOINER Ø6X4MM	2,80 €	1	2,80 €
	TOTAL COST		294,40 €

ELECTRONICS AND PROPULSION	€/UNIT	UNITS	TOTAL
10X6 ELECTRIC EP APC	4,60 €	4	18,40 €
ESC SKYWALKER 50A V2	23,90 €	1	23,90 €
BATTERY RECHARGEABLE NIMH AA 2000MAH 1.2V KAVAN	2,90 €	6	17,40 €
SERVO EMAX ES3004 17GRS/3.5KG	13,90 €	8	111,20 €
CONNECTEUR PK 4MM V2 (3 PAIRES)	1,70 €	1	1,70 €
CABLE SERVO PREMIUM 0.35MM ²	2,50 €	5	12,50 €
FUTABA CONNECTOR (X6)	2,50 €	2	5,00 €
LIPO SLS XTRON 3200MAH 3S 20C	34,50 €	3	103,50 €
LIPO 500MAH 2S 45C	7,10 €	1	7,10 €
EMAX GT2820 985KV	30,50 €	2	61,00 €
FS-I6 + FS-IA6 MODE 2 FLYSKY	85,90 €	1	85,90 €
XT90 MALE+FEMALE	1,90 €	1	1,90 €
XT60 MALE	0,70 €	2	1,40 €
	TOTAL COST		450,90 €

ACCESSORIES	€/UNIT	UNITS	TOTAL
MAIN LANDING GEAR 6CC	20,50 €	1	20,50 €
TAILWHEEL KIT 6CC	6,20 €	1	6,20 €
BOWDEN CABLE M2	3,00€	2	6,00€
LIGHTWEIGHT PLA 0,5KG (WINGLETS, HATCHES)	31,90 €	1	31,90 €
TOUGH PLA 1KG (JOINTS, CONTROL HORNS)	45,90 €	1	45,90 €
SCREWS, NUTS AND WASHERS	6,50 €	1	6,50 €
	TOTAL COST		117,00 €

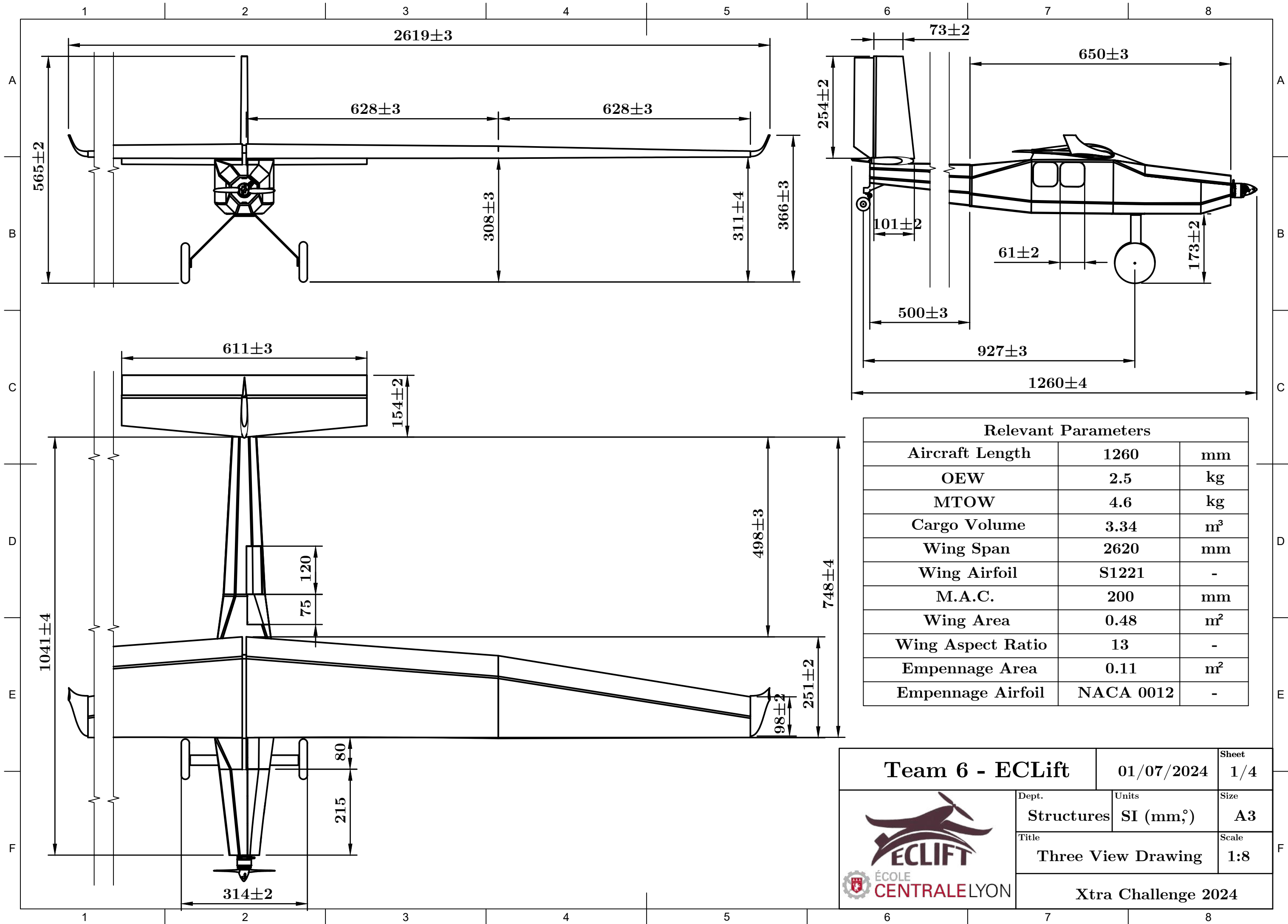
CONSUMABLES	€/UNIT	UNITS	TOTAL
ADHESIVE ALUMINIUM 50MM X 45M	6,50 €	1	6,50 €
HINGE TAPE 50MM / 20M	4,10 €	1	4,10 €
CYANOACRILATE 100ML + ACTIVATOR	24,60 €	1	24,60 €
	TOTAL COST		35,20 €

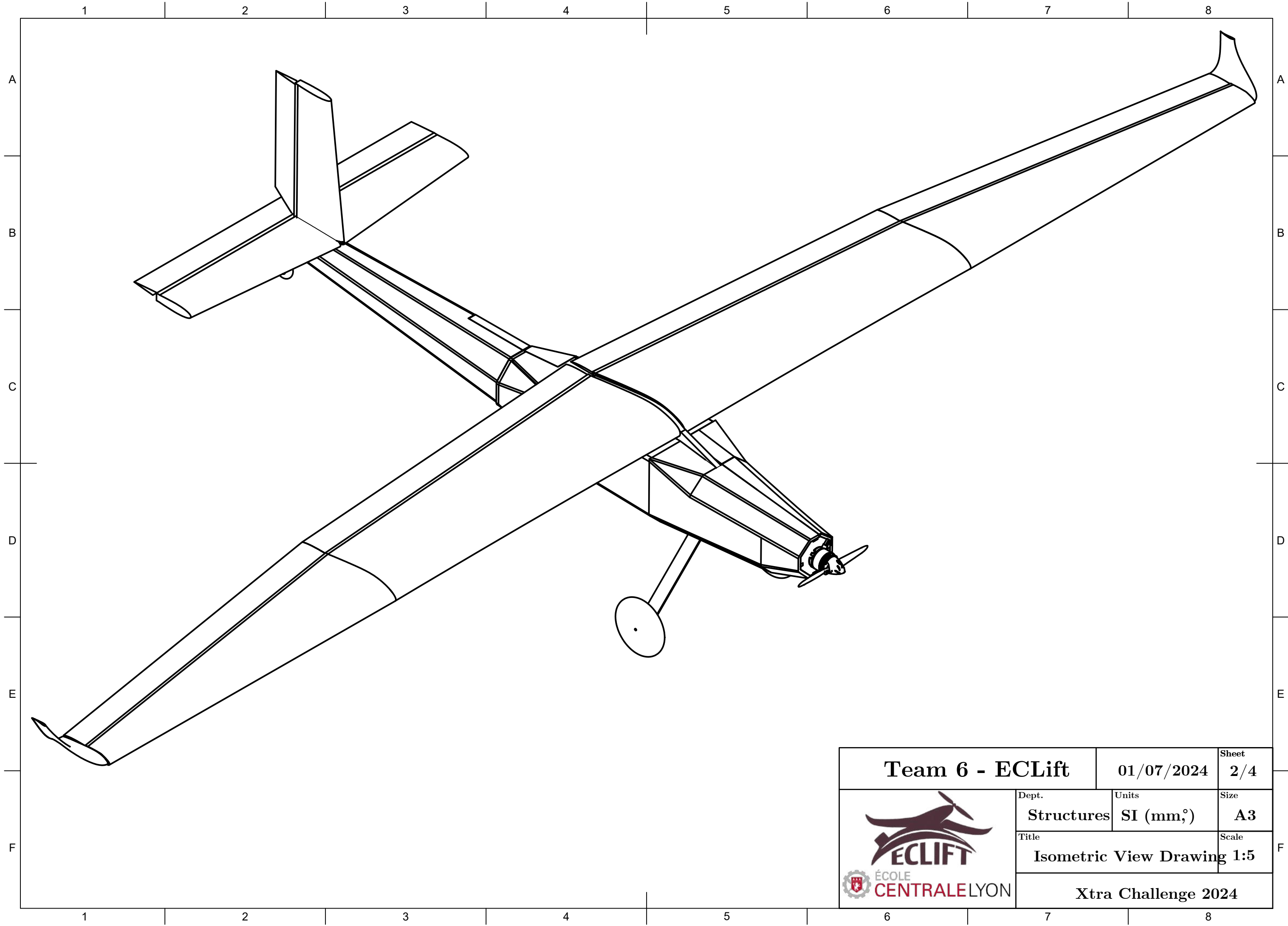
MATERIAL UNIT COST	897,50 €
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PARTICIPATION COSTS	€/UNIT	UNITS	TOTAL
FULL REGISTRATION FEES	150,00 €	6	900,00 €
PARTIAL REGISTRATION FEES	35,00 €	4	140,00 €
TRAVEL	120,00 €	10	1.200,00 €
	TOTAL COST		2.240,00 €

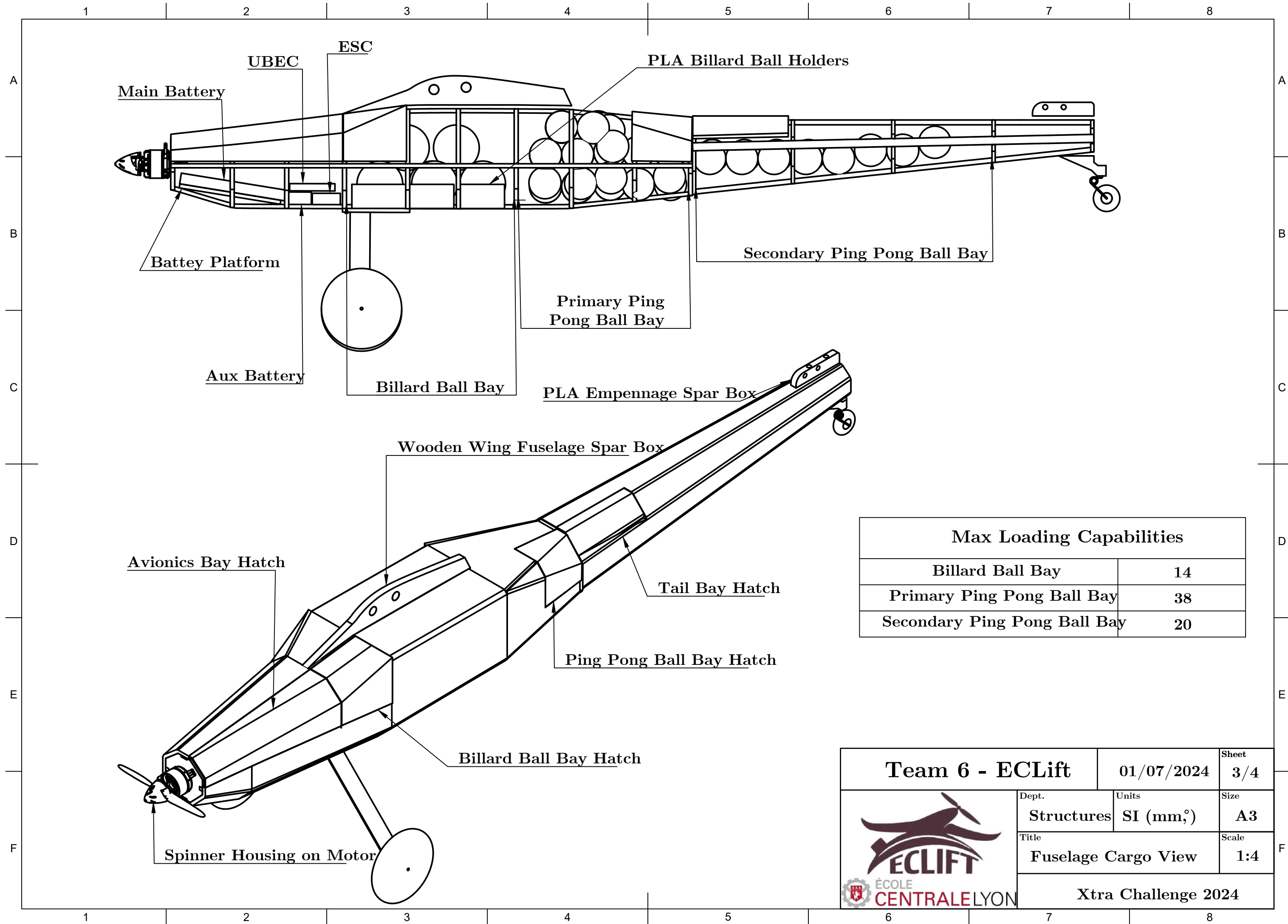
TOTAL PROJECT VALUE	3.137,50 €
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Annex II: Aircraft Drawings

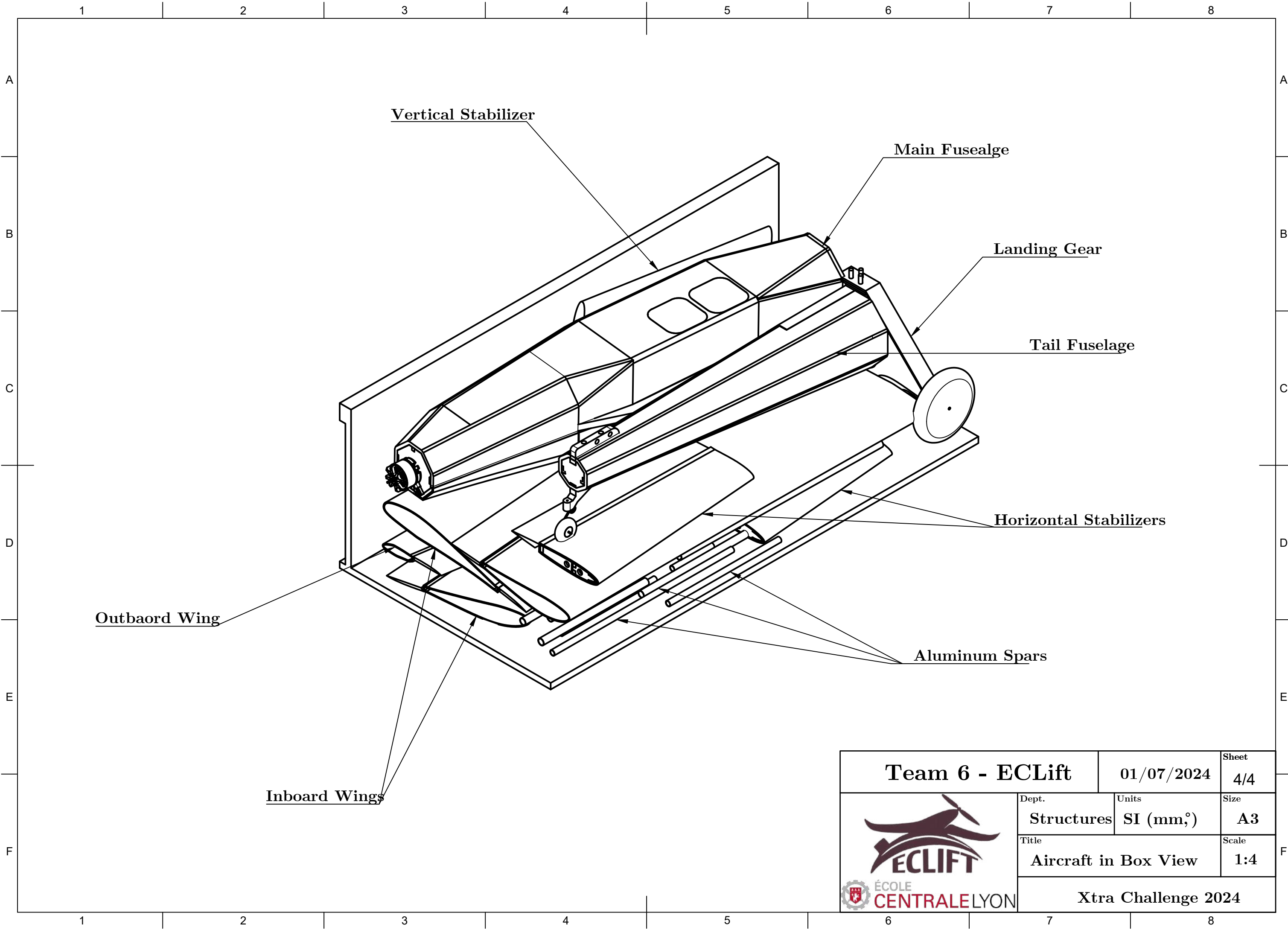




Team 6 - ECLift		01/07/2024	Sheet 2/4
  ÉCOLE CENTRALE LYON	Dept. Structures	Units SI (mm,°)	Size A3
	Title Isometric View Drawing		Scale 1:5
	Xtra Challenge 2024		



Team 6 - ECLift		01/07/2024	Sheet 3/4
  ÉCOLE CENTRALE LYON	Dept.	Units	Size
	Structures	SI (mm,°)	A3
	Title		Scale
	Fuselage Cargo View		1:4
Xtra Challenge 2024			



Team 6 - ECLift		01/07/2024	Sheet 4/4
 	Dept.	Units	Size
	Structures	SI (mm,°)	A3
	Title		Scale
	Aircraft in Box View		1:4
Xtra Challenge 2024			